Zooming In, Zooming Out: Reconciling Treatment Effects across Aggregation Levels*

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Replication package: https://github.com/paulhbeaumont/invariance-replication

Abstract

Are treatment effects estimated at different aggregation levels (e.g., firm-market, firm, and industry) comparable? We show that implicit weights, Jensen's inequality, and entry and exit can generate mechanical differences in estimands. Using simulations, we characterize the conditions under which these differences become meaningful and find that these conditions are met in eight widely used datasets. We develop methods to restore comparability: midpoint growth with midpoint weights and Poisson deliver disaggregation invariance, while standard growth with lagged weights delivers aggregation invariance. A fiscal-multiplier application inspired by Nakamura and Steinsson (2014) illustrates how our methods help distinguish confounding shocks, spillovers, and extensive-margin changes at little additional cost, and how more standard methods yield large biases.

*The views expressed in this paper are those of the authors and do not necessarily represent those of the Federal Reserve Bank of Atlanta or the Federal Reserve system.

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1 Introduction

Empirical researchers often estimate treatment effects at different levels of aggregation. Suppose a researcher studies the effect of a policy designed to stimulate sales among a targeted subset of firms. While the economic object of interest is firm sales, the researcher may have access to more disaggregated data, such as firm sales by product market. An important benefit of moving to a more disaggregated level is that it allows the researcher to include product-market-by-time fixed effects, thereby controlling for product-market shocks that may be correlated with treatment. But does the researcher still estimate the same firm-level treatment effect when moving to a more disaggregated level?

Similar questions arise when aggregating. The researcher may be interested in assessing whether the policy has effects beyond directly treated firms. For instance, treated firms may expand by taking market share away from control firms, leaving aggregate sales unchanged. To assess whether the policy generates spillovers beyond individual firms, a standard approach is to aggregate firm-level data, for instance to the industry level, and regress industry-level sales growth on the share of treated firms. Comparing firm-level and industry-level estimates should then reveal whether the policy generates positive or negative spillovers on other firms, and how large these spillovers are. But are estimates directly comparable across levels of aggregation?

In the first part of the paper, we show that, while the literature routinely treats disaggregated regressions as equivalent to baseline firm-level regressions, standard disaggregated regressions generally identify different population coefficients than the baseline firm-level regression, namely the baseline firm-level treatment effect. This is because standard approaches introduce several deviations from the baseline specification that can generate qualitatively different results. We then develop methods that preserve the baseline population coefficients when moving to more disaggregated data, a property we refer to as "disaggregation invariance." In the second part of the paper, we turn to aggregation and develop methods that allow researchers to draw meaningful conclusions about the presence and magnitude of spillovers from comparisons across levels of aggregation ("aggregation invariance"). Importantly, we show that conventional comparisons can be misleading, even qualitatively, in the absence of these adjustments.

Using eight widely used datasets on trade, employment, syndicated loans, patents, and stock holdings, we first document that disaggregated data share common characteristics that can cause population coefficients to differ across levels of aggregation. First, disaggregation can substantially reweight baseline units because some units contain many more disaggregated observations than others. For instance, in Panjiva–Compustat, a firm at the 75th percentile has positive exports in 10 times more destination–HS6 product cells than a firm at the 25th percentile (Figure 1). Second, entry and exit are pervasive at lower levels of aggregation, so studying changes at a disaggregated

level can drop a substantial share of economic activity that the baseline level encompasses. In Panjiva–Compustat, for example, moving from the firm level to the firm–destination–product level drops 82% of export activity for the median firm over a three-year horizon.

Given these properties of disaggregated data, how does changing the level of aggregation affect treatment effect estimates? We start with disaggregation and ask under which conditions the *disaggregated* regression

$$\Delta S_{im} = \alpha_d + \beta_d \, Treated_i + \epsilon_{im} \tag{1}$$

and the *baseline* regression

$$\Delta S_i = \alpha_b + \beta_b \, Treated_i + \epsilon_i \tag{2}$$

identify the same estimand, that is, $\beta_d = \beta_b$. Throughout the paper, we focus on population regression coefficients, abstracting from sampling variation and finite-sample estimation issues.

For illustration, we take the firm-market as the disaggregated level, the firm as the baseline level, and let S denote sales. In this setting, disaggregation allows ruling out product demand shocks (e.g., Berman, Martin and Mayer, 2012; Paravisini, Rappoport, Schnabl and Wolfenzon, 2014). In practice, S can represent any variable whose value at the baseline level can be exactly decomposed as the sum of its values at the disaggregated level. For instance, moving from firm-level bank credit to firm-bank-level credit allows absorbing credit demand shocks (e.g., Khwaja and Mian, 2008; Iyer, Peydró, da Rocha-Lopes and Schoar, 2014), while moving from area-level employment to industry-area employment allows absorbing industry-level shocks (e.g., Mian and Sufi, 2014; Fonseca and Matray, 2024).

We show that under most standard empirical specifications, the disaggregated and baseline regressions identify different estimands, i.e., $\beta_d \neq \beta_b$, even when they use the exact same specification (e.g., the same fixed effects). Three main reasons drive the discrepancy. First, when observations are equally weighted, firms active in more markets receive greater weight in the firm-market regression, leading to a different treatment effect estimate. Second, even with appropriate weighting, Jensen's inequality implies that the firm-level growth rate and the average firm-market-level growth rate will differ when growth rates are nonlinear in sales levels (e.g., log growth rate). Third, firms may enter or exit markets, generating zeros at the firm-market level and, consequently, missing values when using standard or log growth rates.

We propose two approaches that ensure the disaggregated and baseline regressions identify the same estimand, $\beta_d = \beta_b$ ("disaggregation invariance"). First, using the midpoint growth rate $\Delta S_t = 2 \cdot (S_t - S_{t-h})/(S_t + S_{t-h})$ and weighting firm-market observations by $(S_{imt} + S_{imt-h})/(S_{it} + S_{it-h})$ ("midpoint weights") yields identical population coefficients across aggregation levels. Second, estimating treatment effects on sales levels using Poisson regressions also delivers disaggregation

invariance. Both approaches accommodate zeros at the disaggregated level, which is central to restoring disaggregation invariance. Midpoint weights align firm–market observations with their contribution to firm-level growth. Poisson estimation instead exploits the fact that sales levels aggregate linearly: with firm–market fixed effects in the disaggregated regression, the first-order condition for the treatment effect aggregates to the corresponding firm-level first-order condition.

Using simulations, we characterize when disaggregation invariance breaks under standard empirical specifications. Implicit weights become important when treatment effects are heterogeneous across firms and correlate with the number of markets in which firms operate. Jensen's inequality, in turn, matters when treatment affects the within-firm dispersion of growth rates. Finally, entry and exit only cause disaggregation invariance to break when treatment affects market participation. We then show that midpoint growth rates with midpoint weights preserve disaggregation invariance across all three settings. Moreover, this approach delivers an exact decomposition of firm-level growth into intensive-margin, entry, and exit components, allowing researchers to quantify directly the contribution of each channel to the overall treatment effect.

We then apply our methodology to real-life data through a study of local fiscal multipliers.[1] Adapting the methodology of Nakamura and Steinsson (2014), we ask whether military spending increases local economic activity. Our analysis uses publicly available data, allowing us to make the constructed dataset and replication code publicly available.[2] We first investigate whether industry shocks act as a confounding factor, i.e., whether counties receiving more government spending are exposed to systematically different industry shocks. We show that log growth rates can lead researchers to erroneously conclude that fiscal multipliers are explained away by industry shocks: the point estimate shrinks by 55% and is no longer significant at conventional levels. Using midpoint growth with midpoint weights, we show that including industry–year fixed effects reduces the coefficients by only 15% and that they remain statistically significant at the 5% level.

In the second part of the paper, we turn to aggregation. Let $Share\ Treated_j$ denote the weighted share of treated firms in j. We say that aggregation invariance holds if, in the absence of violations of SUTVA, the *aggregated* regression

$$\Delta S_j = \alpha_a + \beta_a\, Share\ Treated_j + \epsilon_j \tag{3}$$

identifies the same estimand as the baseline firm-level regression with industry fixed effects,

$$\Delta S_i = \alpha_{j(i)} + \beta_b\, Treated_i + \epsilon_i, \tag{4}$$

[1] This question has attracted long-standing interest among researchers in economics (e.g., Blanchard and Perotti, 2002; Auerbach and Gorodnichenko, 2012; Chodorow-Reich, 2019) and finance (e.g., Goldman, 2020; Pinardon-Touati, 2026).

[2] Replication package: https://github.com/paulhbeaumont/invariance-replication.

so that $\beta_a = \beta_b$. For illustration, we take j to denote the industry, $j(i)$ firm i's industry, and α_j the industry fixed effects. Comparing firm- and industry-level coefficients therefore aims to detect SUTVA violations across firms within industries (e.g., Garthwaite, 2014; Helm, 2020). Researchers also aggregate firm-level outcomes to the geographical-area level to detect spillovers across firms within areas (e.g., Huber, 2018; Giroud and Mueller, 2019; Mian, Sarto and Sufi, 2023), and aggregate firm-bank outcomes to the firm level to study credit reallocation across banks within firms (e.g., Khwaja and Mian, 2008; Cingano, Manaresi and Sette, 2016; Beck, Da-Rocha-Lopes and Silva, 2021).

We show that the two approaches proposed to restore disaggregation invariance do not, in general, deliver aggregation invariance. Poisson regressions do not preserve the same log-linear specification after aggregation. Midpoint growth rates would require defining industry treatment exposure with weights affected by treatment, making the regressor depend on the outcome.

Instead, we show that standard growth rates with lagged sales weights achieve aggregation invariance. A technical challenge is that standard growth rates are not defined when firms enter the sample. We accommodate entry by exploiting a decomposition of industry growth into intensive- and extensive-margin components. Weighting firm-level observations by lagged sales shares allows us to recover the incumbent contribution. We then recover the total industry-level effect by separately estimating the contribution of firm entry through a regression of new firms' sales, relative to lagged industry sales, on the industry share of treated firms. This two-step approach also reconciles inference across aggregation levels when standard errors are clustered at the aggregate level.

Why are industry fixed effects needed in the baseline regression when aggregating? We show that without industry fixed effects, the firm-level coefficient already picks up part of the spillover. Since the aggregation approach estimates spillovers by comparing the industry coefficient with the firm-level coefficient, omitting industry fixed effects mechanically understates the magnitude of spillovers in the aligned comparison without entry, especially when most treatment variation occurs across industries. In other words, the aggregation approach compares coefficients identified from across-industry variation with those identified from within-industry variation. An alternative way to perform this decomposition is to include the industry treatment share directly in the firm-level regression (the "exposure-to-treated" approach; see e.g., Crépon, Duflo, Gurgand, Rathelot and Zamora, 2013; Cai and Szeidl, 2024; Lerche, 2025).[3] We show that, under aggregation invariance, the aggregation and exposure-to-treated approaches are equivalent.

Using simulations, we show that aggregation invariance can break for the same reasons as disaggregation invariance: implicit weights, Jensen's inequality, and entry and exit. In particular, we show that spillovers can be hard to detect when treatment also affects firm entry. For in-

[3]This specification is sometimes referred to as the linear-in-means model in the peer effects literature (Manski, 1993).

stance, when treatment encourages firm entry while reducing sales at incumbent firms, the positive contribution of new firms can offset negative spillovers on incumbents. As a result, firm- and industry-level log-growth coefficients can appear similar even when spillovers are substantial. We show that our two-step approach separates these contributions and isolates spillover effects.

While disaggregation invariance does not require taking a stance on whether treatment is random at the firm level (in fact, researchers often disaggregate precisely when they suspect that treatment is not random), more structure is needed when aggregating. First, we impose random treatment at the firm level to simplify the interpretation of spillover effects.[4] Second, we also require industry exposure to be unrelated to treatment responses among treated firms. This second condition is violated if, for instance, industry exposure is correlated with heterogeneity in firm-level response to the treatment.growth

In practice, we recommend a three-step procedure to assess the role of treatment heterogeneity in estimating spillovers: (i) control for cross-industry differences in known sources of treatment heterogeneity; (ii) systematically control for cross-industry differences in firm size, which affect both the measure of treatment exposure and the contribution of treated firms to industry growth; (iii) assess whether, after these controls, industries with different exposure systematically differ in the observable characteristics of treated firms (covariate-balance tests).

Returning to our fiscal multiplier application, we ask whether local government spending has effects beyond directly affected counties. Using standard growth with matched weights, we find that a $1,000 increase in annual government spending per resident corresponds to a 2.7 percentage-point increase in employment growth at the county level, but a 10.8 percentage-point increase at the state level, implying a spillover of 8.1 percentage points. We show that a naive comparison of county- and state-level coefficients understates the spillover magnitude by about one-third. Although exposure to county spending correlates with some observed county characteristics, controlling for the imbalances does not qualitatively alter our conclusions, suggesting a limited role for observed treatment heterogeneity in our spillover estimates.

The remainder of the paper proceeds as follows. Section 2 documents two properties of disaggregated data across eight datasets. Section 3 studies disaggregation: it defines disaggregation invariance, shows why standard specifications break it, and establishes that midpoint growth rates with midpoint weights and Poisson estimation restore it. Section 4 turns to aggregation, where restoring invariance requires aligning the firm- and industry-level population equations and a two-step decomposition that separates incumbent and entry effects, and then addresses treatment-effect heterogeneity through matched controls and balance diagnostics.

[4]Spillover effects can also be defined and estimated with non-random treatment, but identification then requires additional assumptions about treatment selection and exposure (e.g., Barkley, Hudgens, Clemens, Ali and Emch, 2020).

Contribution to the literature. We contribute to a growing methodological literature on the disaggregation and aggregation of reduced-form empirical estimates.

Research on disaggregation was spurred by the increasing availability of detailed microdata, which makes it possible to link aggregate changes to micro-level components. This literature traces its origin to Davis and Haltiwanger (1992), who first show how aggregate employment movements can be decomposed into establishment-level contributions and introduces the midpoint growth rate to accommodate zeros arising from entry and exit, thereby delivering exact decompositions of aggregate changes. Similar methodologies have been applied in a wide range of settings to trace the micro origins of aggregate moments.[5]

By contrast, relatively less attention has been paid to whether and how changes in aggregation levels affect treatment effect estimates. Notable exceptions include Moulton (1990), who discusses inference issues when regressing disaggregate outcomes on aggregate ones. Related to the treatment of zeros in disaggregated data, Cohn, Liu and Wardlaw (2022) and Chen and Roth (2024) show that common transformations used to deal with zeros can yield reduced-form estimates that are biased or lack a clear economic interpretation.[6] Degryse, De Jonghe, Jakovljević, Mulier and Schepens (2019) show that the use of high-dimensional fixed effects in disaggregated data can induce selection and bias reduced-form estimates. Finally, Beaumont, Libert and Hurlin (2019) show that using midpoint growth rates with weighted least squares facilitates the incorporation of the extensive margin in the joint estimation of bank and firm fixed effects (Amiti and Weinstein, 2018), an approach subsequently adapted by Fonseca and Matray (2024) and Matray, Müller, Xu and Kabir (2024) to obtain estimates that are consistent across levels of disaggregation. Our paper contributes to this strand of research by formally characterizing the methodological issues raised by estimating treatment effects with disaggregated data, illustrating when discrepancies across levels of disaggregation arise, and proposing simple solutions to restore disaggregation invariance.

As micro-level estimates of causal responses have become more available (Angrist and Pischke, 2010), a substantial body of work, summarized in Nakamura and Steinsson (2018), has developed methods to map reduced-form estimates into aggregate outcomes. Our paper relates specifically to the common practice of comparing estimates across aggregation levels to infer general equilibrium or spillover effects (e.g., Khwaja and Mian, 2008; Huber, 2018; Giroud and Mueller, 2019; Mian, Sarto and Sufi, 2023).[7] In this spirit, Mian, Sarto and Sufi (2023) contrast within- and across-region

[5] See e.g., Baily, Hulten, Campbell, Bresnahan and Caves (1992); Olley and Pakes (1996); Foster, Haltiwanger and Krizan (2001); Hsieh and Klenow (2009); Gabaix (2011); Di Giovanni, Levchenko and Mejean (2014, 2017); Fitzgerald and Haller (2018); Boualam and Mazet-Sonilhac (2025).

[6] Silva and Tenreyro (2006) show that, in the presence of heteroskedasticity, OLS estimates of log-linearized models are necessarily biased due to Jensen's inequality and recommend Poisson specifications instead.

[7] Alternative approaches include deriving conditions under which cross-sectional estimates can inform general equilibrium effects (e.g., Chodorow-Reich, 2019; McKay and Wolf, 2023; Wolf, 2023), combining multiple cross-sectional estimates to recover aggregate effects (e.g., Arkolakis, Costinot and Rodríguez-Clare, 2012; Guren, McKay, Nakamura and Steinsson, 2021; Sraer and Thesmar, 2023), or constructing general equilibrium models that target

estimates to isolate regional amplification mechanisms of credit shocks, while Herreno (2023) use a structural framework to map relative cross-sectional estimates of bank credit shocks into aggregate elasticities. Relatedly, Ferracci, Jolivet and van den Berg (2014) develop a two-step framework that identifies direct and indirect treatment effects by exploiting variation in individual treatment status and in market-level treatment shares. At the disaggregated level, Berg, Reisinger and Streitz (2021) show that ignoring spillovers can severely bias treatment effect estimates, while Huber (2023) provides econometric guidance for estimating spillover effects in the presence of nonlinearities, measurement error, and multiple spillover channels. Chemla and Hennessy (2026) show how moral hazard can undermine standard tests for SUTVA violations when firms gather information before deciding whether to take up a treatment. In contrast, we show that, even in settings without SUTVA violations or additional frictions, population coefficients generally differ across aggregation levels under standard specifications, casting doubt on the ability of aggregation-based SUTVA diagnostics to avoid false positives. We characterize the sources of these discrepancies across aggregation levels and propose simple approaches to restore aggregation invariance.

Finally, our paper builds on a growing literature that seeks to provide actionable solutions to underappreciated econometric challenges (e.g., Bertrand, Duflo and Mullainathan, 2004; Petersen, 2008; Gormley and Matsa, 2013; Jiang, 2017; Goldsmith-Pinkham, Sorkin and Swift, 2020; Borusyak, Hull and Jaravel, 2021; Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021; Baker, Larcker and Wang, 2022; Cohn, Liu and Wardlaw, 2022; Borusyak and Hull, 2023; Chen and Roth, 2024). We contribute to this literature by introducing the notion of disaggregation and aggregation invariance, which formalizes the common intuition that, in the absence of compositional or general equilibrium effects, treatment effects should not depend on the level of aggregation. We show, using simulations and real data, that traditional methods do not deliver disaggregation or aggregation invariance, highlighting the need to revisit common rules of thumb in applied work.

2 Descriptive statistics on disaggregated data

Researchers commonly have access to data that allow them to observe their object of interest, such as firm-level employment, at a more disaggregated level, such as employment by firm and occupation. The benefits of moving the analysis to the disaggregated level are clear: more information and the possibility of including additional fixed effects. But what are the potential costs?

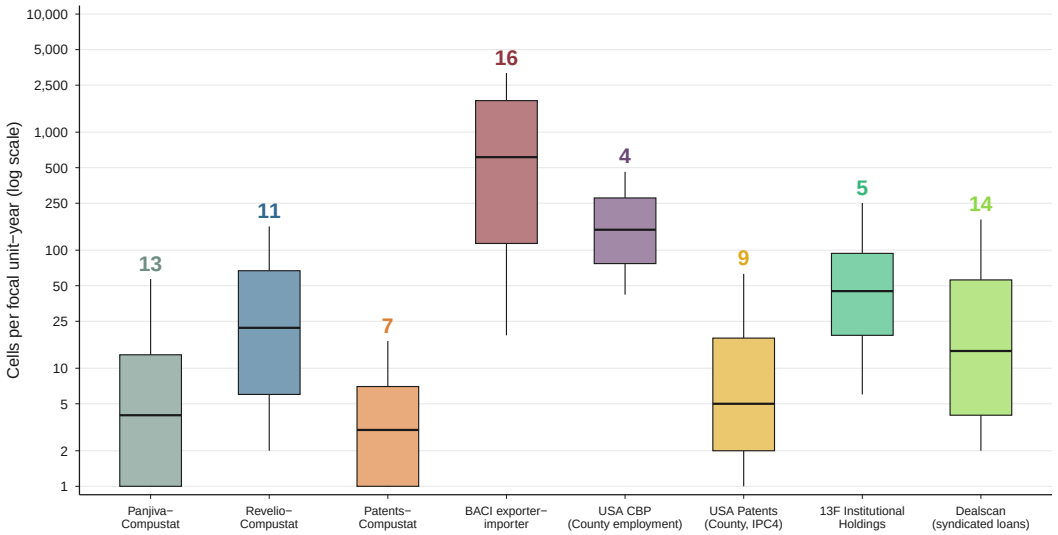
We start by documenting two general characteristics of disaggregated datasets. The analysis draws on several datasets widely used in economics and finance: (i) U.S. export data from Panjiva–Compustat; (ii) U.S. employment data from Revelio–Compustat; (iii) patent data linked

reduced-form moments to assess aggregate effects (e.g., Chodorow-Reich, 2014; Catherine, Chaney, Huang, Sraer and Thesmar, 2022).

to Compustat firms; (iv) the BACI bilateral trade database; (v) U.S. county-level employment data from the U.S. Census Bureau; (vi) U.S. patent data at the county level; (vii) syndicated loan data from Dealscan; and (viii) institutional investor stock holdings from 13F filings.

For each dataset, we define a baseline level and a disaggregated level, with the latter nested within the former. We impose two common restrictions across datasets. First, baseline units must be observed in at least two periods, so that the analysis focuses on continuing units. Second, observations with missing or non-positive values ("inactive observations") are dropped before aggregation. Appendix A describes the levels used in each dataset and the corresponding construction steps.

Figure 1: Number of disaggregated observations per baseline unit across datasets



Note: The figure reports the distribution of the number of disaggregated observations per baseline unit. We compute this number separately by year and then pool baseline observations across years. The middle line denotes the median, boxes span the 25th to the 75th percentile, and whiskers span the 10th to the 90th percentile. The number above each box indicates how many more disaggregated observations a baseline unit at the 75th percentile has relative to a baseline unit at the 25th percentile. In the Panjiva–Compustat data, for example, a firm at the 75th percentile has positive exports in 10 times more destination–HS6 product observations than a firm at the 25th percentile. The aggregation levels are defined as follows, with the baseline level listed first and the disaggregated level listed second: (i) *Panjiva–Compustat:* firm and firm × destination × product (HS6); (ii) *Revelio–Compustat:* firm and firm × occupation; (iii) *Patents–Compustat:* firm and firm × technology class; (iv) *BACI bilateral trade data:* exporter–importer pair and exporter–importer pair × product (HS6); (v) *U.S. county-level employment data:* county and county × NAICS6 industry; (vi) *U.S. patent data:* county and county × IPC4 technology class; (vii) *13F institutional holdings:* fund manager and fund manager × stock; and (viii) *Dealscan:* bank and bank × SIC4 industry.

A first recurring feature of disaggregated datasets is that baseline units differ in the number of observations they contain at the disaggregated level. As a result, changing the level of observation changes the weight placed on different baseline units and can therefore change the estimand. Figure 1 reports the distribution of the number of disaggregated observations per baseline unit across datasets. The figure also reports the ratio of the 75th percentile to the 25th percentile above each box.

We find substantial variation across baseline units in the number of observations at the dis-

aggregated level. In the Panjiva–Compustat data, for example, a firm at the 75th percentile has positive exports in 10 times more destination–HS6 product observations than a firm at the 25th percentile. Similar patterns appear in other firm-level datasets: firms differ substantially in the number of occupations or patent classes they contain. The gap is especially pronounced in BACI, where exporter–importer pairs differ sharply in the number of products traded, and in Dealscan, where banks differ substantially in their number of syndicated loan borrowers. Dispersion is smaller but still meaningful in more aggregated datasets such as U.S. county-level employment and 13F institutional holdings.

Overall, the evidence shows that a disaggregated cell-level specification mechanically assigns more weight to baseline units with many active disaggregated observations. Without explicit reweighting, changing the level of observation therefore changes the relative importance of baseline units in the estimated object.

A second characteristic of disaggregated datasets is the pervasive importance of entry and exit at lower levels of aggregation. Even when a growth rate is well defined at the baseline level, it may be undefined at the disaggregated level because some observations are nonzero in only one of the two periods. As a result, changing the level of aggregation can change the estimand mechanically: growth rates that require positive values in both periods drop entry and exit observations, which breaks the link between growth rates across levels of aggregation.

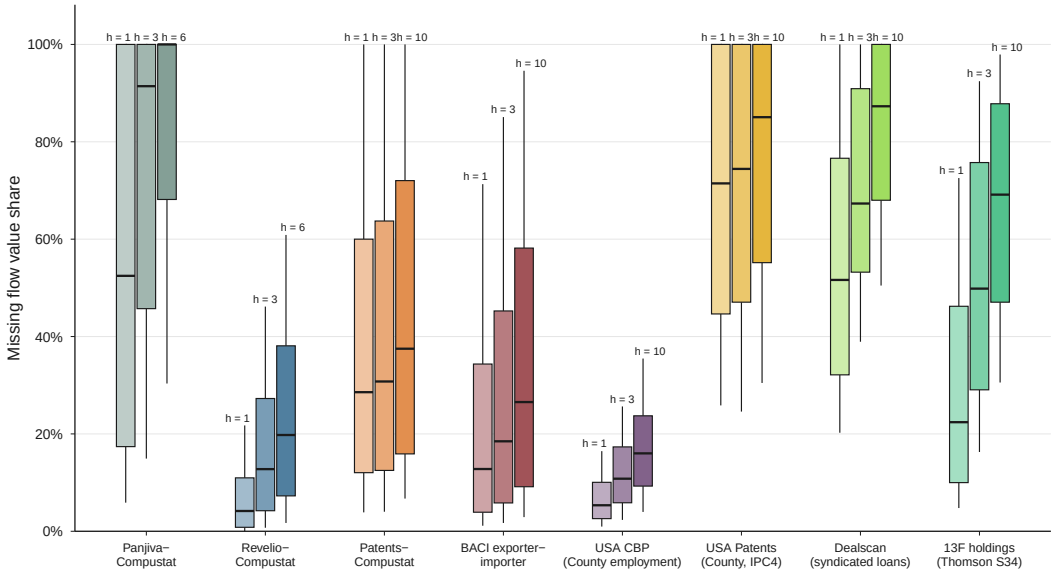
We illustrate the magnitude of this problem by measuring how much baseline-unit activity is lost when entry or exit observations are dropped. Specifically, we define the "extensive-margin share" as

$$\text{Extensive margin share}_{i,t} = \frac{\text{Entry value}_{i,t} + \text{Exit value}_{i,t-h}}{S_{i,t} + S_{i,t-h}}. \tag{5}$$

This measure ranges from 0 to 1 and gives the fraction of the baseline unit's value that a specification restricted to positive disaggregated observations in both periods would drop. A value of 0 implies that all of the unit's value comes from observations present in both $t-h$ and t, so nothing is lost. A value of 1 implies that the unit reshuffles its portfolio entirely, so a regression capturing only the intensive margin would miss 100% of its growth. We restrict attention to baseline units with $S_{i,t-h} > 0$ and $S_{i,t} > 0$, so that the baseline unit is present in both periods. The entry value is the value at t of disaggregated observations that are inactive at $t-h$ but active at t. The exit value is the value at $t-h$ of disaggregated observations that are active at $t-h$ but inactive at t.

Figure 2 reports the results. Three patterns emerge. First, even over short horizons, disaggregation can lead researchers to drop a substantial share of a baseline unit's activity. In Panjiva–Compustat, for example, the median extensive-margin share is 45% at the one-year horizon, with a 75th percentile above 95%. Second, the importance of the extensive margin depends on the nature of the underlying economic activity. Entry and exit are more important, for instance, in stock

Figure 2: Extensive margin share across datasets



Note: The figure reports the distribution of the extensive margin share, as defined by equation (5), across datasets. A value of 30% means that a specification requiring positive disaggregated observations at the beginning and end of the period drops 30% of the baseline unit's activity in value. The middle line denotes the median, boxes span the 25th to the 75th percentile, and whiskers span the 10th to the 90th percentile. We compute the extensive margin share for a short, medium, and long horizon h. The short horizon is one year, the medium horizon is three years, and the long horizon is ten years when possible, or six years otherwise. We compute the shares separately by year and then pool baseline observations across years. The aggregation levels are defined as follows, with the baseline level listed first and the disaggregated level listed second: (i) *Panjiva–Compustat:* firm and firm × destination × product (HS6); (ii) *Revelio–Compustat:* firm and firm × occupation; (iii) *Patents–Compustat:* firm and firm × technology class; (iv) *BACI bilateral trade data:* exporter–importer pair and exporter–importer pair × product (HS6); (v) *U.S. county-level employment data:* county and county × NAICS6 industry; (vi) *U.S. patent data:* county and county × IPC4 technology class; (vii) *13F institutional holdings:* fund manager and fund manager × stock; and (viii) *Dealscan:* bank and bank × SIC4 industry.

portfolios or firm-level export flows than in firms' workforce composition or countries' product trade flows. Third, the extensive margin becomes more important as the horizon increases. In Revelio–Compustat, the median extensive-margin share is 5% at the one-year horizon but reaches 20% at the six-year horizon. Longer differences give disaggregated cells more time to enter or exit, so specifications based only on continuing cells increasingly restrict attention to a selected subset of the baseline unit's activity.

Our analysis suggests that disaggregated datasets have specific characteristics that can cause average treatment effect estimates to substantially differ across levels of aggregation. In the next two sections, we formally study when, and to what extent, average treatment effect estimates diverge across levels, focusing first on disaggregation and then on aggregation.

3 Disaggregation

3.1 Disaggregation invariance

We now formalize the notion of disaggregation invariance. The baseline outcome is firm-level sales, S_{it}, which can be exactly decomposed into firm–market components:

$$S_{it} = \sum_m S_{imt}.$$

Throughout the paper, we consider three commonly used growth-rate definitions: log growth, standard growth, and midpoint growth. For any horizon $h \geq 1$, these growth rates are defined as

$$\Delta^l S_t = \log S_t - \log S_{t-h}, \qquad \Delta^s S_t = \frac{S_t - S_{t-h}}{S_{t-h}}, \qquad \Delta^{mp} S_t = 2\,\frac{S_t - S_{t-h}}{S_t + S_{t-h}}.$$

We omit the horizon h from the growth operators to simplify notation. Disaggregation consists in replacing $\{S_{it}\}_i$ by $\{S_{imt}\}_{i,m}$ in equation (2). We say that a specification achieves disaggregation invariance if the disaggregated and baseline regressions target the same estimand, $\beta_d = \beta_b$, in the following regressions at the firm level and at the firm–market level, which include the same fixed effects:

$$\Delta S_{it} = \alpha_t + \beta_b \, Treated_i + \epsilon_{it}, \tag{6}$$

$$\Delta S_{imt} = \alpha_t + \beta_d \, Treated_i + \epsilon_{imt}. \tag{7}$$

To study the biases associated with common practices in empirical research, we focus on the comparison between the firm-level OLS coefficient and the firm–market OLS coefficient, using log growth as the outcome in both regressions. Treatment is defined at the firm level, so all markets of a given firm share the same treatment indicator. Our analysis compares population coefficients: disaggregation can change the estimand even as the sample size grows arbitrarily large. The discrepancies we study therefore reflect differences in what the regressions estimate, rather than sampling variation.

Importantly, our definition of disaggregation invariance does not require any assumption on treatment being randomly assigned at the firm or firm–market level. In practice, researchers often disaggregate precisely because they suspect that treatment is not random at the firm level, for instance because treated and control firms operate in systematically different markets. In such cases, neither the baseline nor the disaggregated coefficient necessarily admits a causal interpretation. Disaggregation invariance is nonetheless central: it ensures that introducing market-level controls does not mechanically change the estimand through the change in aggregation level. As a result,

it becomes possible to interpret differences between baseline and disaggregated coefficients as reflecting the role of market shocks, rather than as artifacts of disaggregation. When disaggregation invariance fails, estimates obtained with market fixed effects may differ from firm-level estimates even if market shocks are irrelevant, making it impossible to assess whether disaggregation genuinely sharpens identification.

3.2 Why disaggregation invariance breaks

Disaggregation can change the estimand through three channels: implicit weights, Jensen's inequality, and entry and exit. We study these channels in the next three subsections.

3.2.1 Implicit weights

A first source of discrepancy between baseline and disaggregated coefficients is that the number of observations at the disaggregated level varies across firms. As we document in Section 2, some firms operate in many more markets than others. A firm-level OLS regression gives each firm the same weight. An equal-observation firm–market regression instead gives more weight to firms that operate in more markets.

To describe the role of weights, we first set aside entry and exit. We compare the equal-observation firm–market coefficient, $\beta_{d,OLS}$, with the WLS coefficient, $\beta_{d,w}$, using each market's share of the firm's lagged sales,

$$w_{im}^{l} = \frac{S_{im,t-h}}{S_{i,t-h}}.$$

These weights sum to one within each firm, therefore giving each firm the same weight. The weighted regression compares treated and control firms' sales-weighted average market log growth. We discuss the distinction between average market log growth and log growth in total firm sales in the next subsection.

The change in weights does two things. First, it removes the extra weight that OLS gives to firms operating in more markets. Second, within each firm, it replaces equal market weights with sales shares. Let n_i denote the number of markets and $g_{im} = \Delta^{l} S_{imt}$ market log growth. Write

$$\bar{g}_i = \frac{1}{n_i} \sum_{m} g_{im}, \qquad a_i = \sum_{m} w_{im}^{l} g_{im} - \bar{g}_i.$$

Here, $\bar{g}_i$ is average market log growth and a_i is the change in that average when larger markets receive more weight. With an intercept and a binary treatment indicator, the coefficient difference

is

$$\begin{aligned}\beta_{d,w} - \beta_{d,OLS} = E[a_i \mid Treated = 1] - E[a_i \mid Treated = 0] \\ - \left(\frac{\text{Cov}(n_i, \bar{g}_i \mid Treated = 1)}{E[n_i \mid Treated = 1]} - \frac{\text{Cov}(n_i, \bar{g}_i \mid Treated = 0)}{E[n_i \mid Treated = 0]} \right).\end{aligned} \tag{8}$$

The first term matters when giving more weight to larger markets changes average growth differently for treated and control firms. For example, this can happen if treatment raises sales more strongly in a firm's larger markets.

The second term depends on the covariance between the number of markets and average market log growth, relative to the average number of markets in each group. It matters when this relationship differs between treated and control firms. For example, larger firms may operate in more markets and respond more strongly to treatment. We use this example in Section 3.3.2.

3.2.2 Jensen's inequality

A second source of discrepancy is that the logarithm is nonlinear. Because of Jensen's inequality, log growth in total firm sales differs from the sales-share-weighted average of log growth across the firm's markets. This difference can in turn affect the comparison between baseline and disaggregated coefficients. To see why, keep the set of markets fixed and use the lagged sales shares w_{im}^l introduced above. The gap between firm growth and sales-share-weighted market growth is

$$\begin{aligned}J_{it} &\equiv \Delta^l S_{it} - \sum_m w_{im}^l \Delta^l S_{imt} \\ &= \log\left(\sum_m w_{im}^l e^{\Delta^l S_{imt}} \right) - \sum_m w_{im}^l \Delta^l S_{imt} \;\approx\; \frac{1}{2} \text{Var}_{w^l}(\Delta^l S_{imt}).\end{aligned} \tag{9}$$

The equality is exact. The final, second-order approximation is useful for interpreting it. The gap is nonnegative for each firm and approximately rises with the within-firm dispersion of market growth. Recall that $\beta_{d,w}$ is the coefficient from the firm–market regression weighted by these lagged sales shares. Equation (9) then implies

$$\beta_b - \beta_{d,w} = E[J_{it} \mid Treated = 1] - E[J_{it} \mid Treated = 0]. \tag{10}$$

Thus, Jensen's inequality affects the coefficient gap when the average Jensen term differs between treated and control firms. The approximation helps explain when this happens. For example, treatment may increase sales sharply in some of a firm's markets but have little effect in others. If this raises the within-firm dispersion of market growth among treated firms relative to control firms, the Jensen term will tend to be larger for treated firms. Firm-level log growth then captures a change in the dispersion of market growth in addition to its weighted average.

3.2.3 Entry and exit

Log growth at the firm–market level is defined only where a firm sells in both periods, so it omits entry into new markets and exit from old ones. Let S_{it}^c be the firm's sales in continuing markets. Among firms with at least one such market, let e_{it} be the share of current sales from new markets and x_{it} the share of lagged sales from markets the firm has left. By definition, continuing-market sales satisfy

$$S_{it}^c = (1 - e_{it})S_{it}, \qquad S_{i,t-h}^c = (1 - x_{it})S_{i,t-h}.$$

Dividing these identities and rearranging gives

$$\frac{S_{it}}{S_{i,t-h}} = \frac{S_{it}^c}{S_{i,t-h}^c}\frac{1 - x_{it}}{1 - e_{it}}.$$

Taking logs yields

$$\Delta^l S_{it} = \Delta^l S_{it}^c + \log(1 - x_{it}) - \log(1 - e_{it}). \tag{11}$$

Let $\beta_{b,c}$ be the firm-level coefficient using log growth in continuing-market sales, estimated on the same firms as β_b. Taking treated–control differences gives the exact extensive-margin contribution:

$$\begin{aligned}\beta_b - \beta_{b,c} = {} & E[\log(1 - x_{it}) - \log(1 - e_{it}) \mid Treated = 1] \\ & - E[\log(1 - x_{it}) - \log(1 - e_{it}) \mid Treated = 0].\end{aligned} \tag{12}$$

The contribution is zero when the combined entry–exit term has the same expectation in the two groups, or when entry and exit offset each other.

3.2.4 Putting the three channels together

We now combine the three channels for log growth, using the same firms with at least one continuing market throughout. Let $\beta_{d,OLS}$ be the equal-observation coefficient on continuing firm–market observations. As in Section 3.2.2, $\beta_{d,w}$ uses sales shares, now calculated relative to each firm's lagged continuing-market sales. Let J_{it}^c be the Jensen term in equation (9) calculated among those markets. All regressions include an intercept and the same binary treatment indicator, without additional controls. As we prove in Appendix D.1, the coefficient gap is exactly

$$\begin{aligned}\beta_b - \beta_{d,OLS} = {} & \underbrace{(\beta_{d,w} - \beta_{d,OLS})}_{\text{Implicit weights}} + \underbrace{(E[J_{it}^c \mid Treated = 1] - E[J_{it}^c \mid Treated = 0])}_{\text{Jensen}} \\ & + \underbrace{(\beta_b - \beta_{b,c})}_{\text{Entry and exit}}.\end{aligned} \tag{13}$$

3.2.5 Restoring disaggregation invariance

This section discusses two approaches that restore disaggregation invariance, i.e., ensure that the disaggregated and baseline regressions identify the same estimand, $\beta_d = \beta_b$, in the absence of market-level shocks: midpoint growth rates and Poisson estimation.

Proposition 1 (Disaggregation invariance)**.** *The disaggregated and baseline regressions identify the same estimand, $\beta_d = \beta_b$, under either specification on the same firms: (i) midpoint growth rates with equal weights at the firm level and weighted least squares with midpoint weights $w_{im}^{mp} = (S_{imt} + S_{im,t-h})/(S_{it} + S_{i,t-h})$ at the firm–market level; or (ii) the Poisson level specifications (15) and (16) with firm and firm–market fixed effects. Both accommodate firm–market entry and exit.*

Midpoint growth rates. The midpoint growth rate simultaneously addresses the three sources of non-invariance identified above: weighting, nonlinearity, and entry and exit. Two features of the midpoint growth rate are key. First, aggregation is linear: firm-level midpoint growth is a weighted average of firm–market midpoint growth rates. As a result, Jensen's inequality does not arise, in contrast to log growth rates. Second, the midpoint growth rate is well defined in the presence of zeros, ensuring that firm–market entry and exit are incorporated symmetrically. Firm-level growth therefore aggregates changes over the same domain as the firm–market data. We formally show that midpoint growth rates with midpoint weights satisfy disaggregation invariance in Appendix D.3.

It also separates growth into extensive and continuing-market components. Let E_{it} be current sales in new markets and X_{it} lagged sales in markets the firm has left. Then

$$\Delta^{mp} S_{it} = \underbrace{\frac{2E_{it}}{S_{it} + S_{i,t-h}}}_{\text{Entry}} - \underbrace{\frac{2X_{it}}{S_{it} + S_{i,t-h}}}_{\text{Exit}} + \underbrace{\frac{2(S_{it}^c - S_{i,t-h}^c)}{S_{it} + S_{i,t-h}}}_{\text{Continuing markets}}. \tag{14}$$

A natural concern is that midpoint weights depend on contemporaneous sales, which may themselves be affected by treatment. We discuss this issue in Appendix C. We show that midpoint weights introduce no additional identification problem when treatment is randomly assigned at the firm level: in that case, the midpoint-weighted firm–market regression targets the same estimand as the firm-level regression. When identification instead relies on firm–market comparisons conditional on market fixed effects, midpoint weights can create bias only if they differentially weight idiosyncratic firm–market shocks for treated and control firms within the same market. We show that across the eight datasets used in Section 2, midpoint weights are strongly correlated with standard weights , suggesting that firm–market innovations play a limited role in shaping midpoint weights.

In practice, a simple robustness check is to replace midpoint growth rates with standard firm–

market growth rates and use predetermined weights $w^s_{im} = S_{im,t-h}/S_{i,t-h}$. As we explain in Section 4.3, this specification identifies the effect of treatment on sales growth in markets in which the firm was already active. The contribution of treatment to market entry can then be recovered separately using a firm-level regression of entry rates on treatment. While this alternative does not incorporate intensive- and extensive-margin adjustments within a single regression, it provides a transparent robustness check based on predetermined weights while still allowing comparisons across aggregation levels.

Poisson estimation. Poisson estimation provides an alternative route to disaggregation invariance that does not rely on growth rates or on weights constructed from post-treatment outcomes. Instead, it models sales directly in levels and exploits the multiplicative structure of the conditional mean. Consider a two-period panel with one pre-treatment and one post-treatment observation. At the firm level, we estimate the Poisson regression

$$E(S_{it} \mid i,t) = \exp(\alpha_i + \mu_t + \beta_b \, Treated_i \times Post_t)\,, \tag{15}$$

where α_i is a firm fixed effect and μ_t a time fixed effect. At the firm–market level, we estimate

$$E(S_{imt} \mid i,m,t) = \exp(\alpha_{im} + \mu_t + \beta_d \, Treated_i \times Post_t)\,, \tag{16}$$

where α_{im} is a firm–market fixed effect.

Appendix D.2 shows that the first-order condition for β_d in the disaggregated regression is identical to the first-order condition for β_b in the baseline regression. As a result, the two specifications target the same estimand ($\beta_d = \beta_b$).

This equivalence follows from the multiplicative structure of the Poisson model. In the fitted mean, treatment enters through a common multiplier across the markets of a given firm. Firm–market fixed effects absorb all permanent differences in sales across markets, and the first-order condition aggregates naturally over markets. The treatment coefficient is therefore unchanged when sales are summed across markets, yielding disaggregation invariance.

Inference. Our definition of disaggregation invariance involves an equality of the baseline and disaggregated estimand. What about standard errors? In Appendix D.4, we show that, under the midpoint-growth specification, clustering standard errors at the firm level yields identical standard errors in the firm-level and firm–market-level regressions. Thus, our approach reconciles both treatment effect estimates and inference across aggregation levels.

3.3 When does disaggregation invariance matter?

Disaggregation invariance is primarily a concern when the discrepancy between baseline and disaggregated coefficients is economically meaningful. In certain applications, differences between firm-level and firm–market estimates may be small, in which case standard empirical practices remain a good approximation. In this section, we use simulations to quantify when disaggregation invariance is empirically important and to clarify the mechanisms that break disaggregation invariance.

3.3.1 Data-generating process

To quantify the gap between firm-level and firm–market estimates under standard empirical specifications, we turn to simulations. We describe below how we generate firm-market sales and compare treatment coefficients across levels of aggregation and growth-rate specifications. The detailed data-generating process, baseline parameters, and implementation are provided in Appendix E.1.

Our data-generating process builds on the framework of Eaton, Kortum and Kramarz (2011). Firms sell across multiple markets in two periods, and both market entry and sales conditional on entry depend on firm productivity. Treatment is modeled as a positive productivity shock, affecting sales through both the intensive margin (higher sales in existing markets) and the extensive margin (entry into additional markets and reduced exit). Treatment is randomly assigned at the firm level, independently of initial productivity and all other firm characteristics.

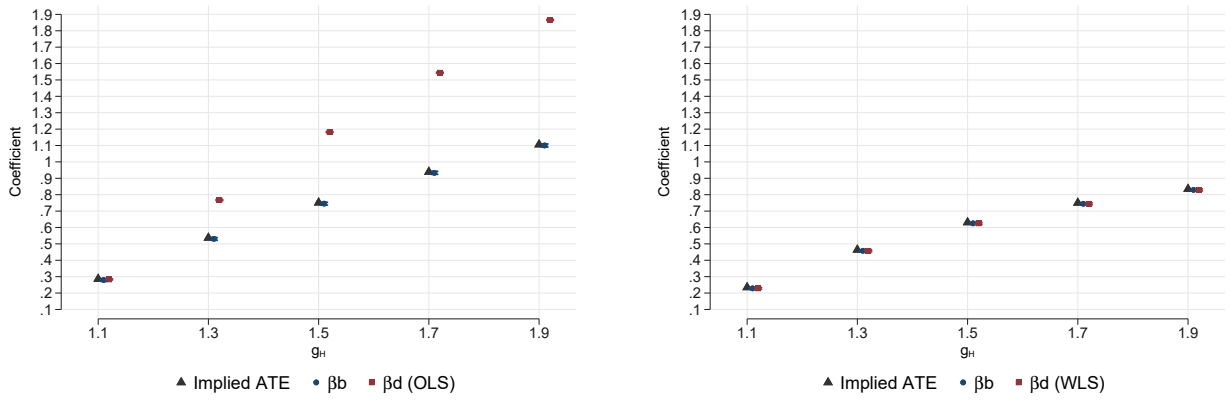
A key advantage of this framework is that it generates sales in levels in both periods. We can therefore apply different growth-rate transformations to the same simulated economy and compare the resulting regression coefficients with the firm-level average treatment effect implied by the data-generating process.

For each source of disaggregation-invariance failure—implicit weights, Jensen's inequality, and entry and exit—we compare log growth estimated by OLS with midpoint growth estimated using midpoint weights. The former reflects common empirical practice, while the latter is our proposed solution. In Appendix E, we also compare midpoint growth with midpoint weights to alternative specifications designed to isolate the specific source of departure from disaggregation invariance. Together, these comparisons establish the conditions under which standard specifications generate economically meaningful departures from disaggregation invariance.

3.3.2 Implicit weights

As discussed in Section 3.2.1, giving greater weight to firms operating in more markets matters when treatment effects are heterogeneous and systematically vary with the number of markets. In our simulation, we capture this mechanism by distinguishing between two types of firms: high-

Figure 3: Disaggregation invariance: Implicit weights



(a) Log growth rate – OLS

(b) Midpoint growth rate – WLS

Note. Firms have low or high initial productivity with equal probability. High-productivity firms operate in more markets, with market counts fixed across periods. The x-axis gives their productivity multiplier under treatment, g_H; the multiplier for low-productivity treated firms is fixed at $g_L = 1.1$. Control firms experience no productivity change. Bars show 95% confidence intervals for mean estimates across simulation repetitions. Appendix E.1 gives the parameter values and implementation details.

productivity firms, which operate in more markets and respond more strongly to treatment, and low-productivity firms, which operate in fewer markets and respond less. To isolate the role of implicit weights, we hold each firm's number of markets fixed across periods, ruling out entry and exit.

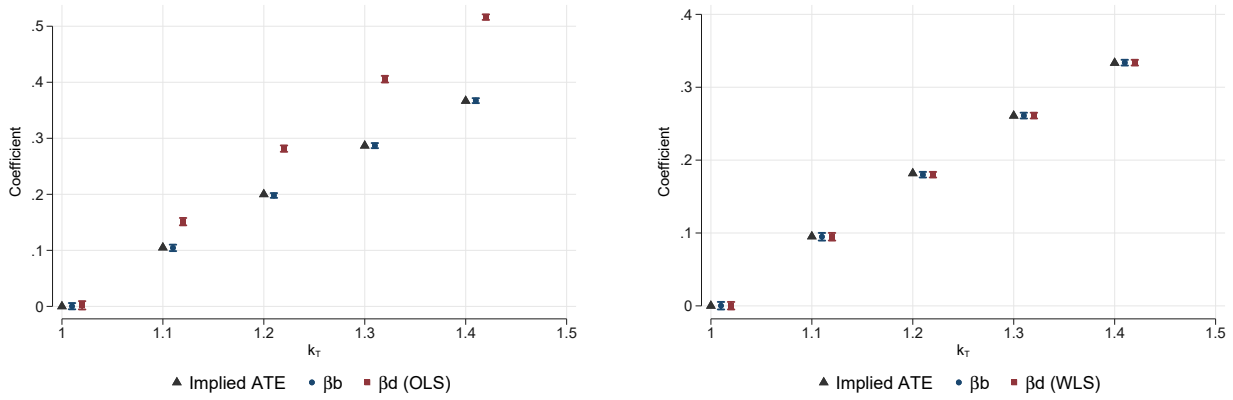
Because high-productivity firms operate in more markets, the firm–market coefficient over-represents the effect for these firms, whereas the firm-level coefficient assigns equal weight to each firm. Figure 3 varies the productivity multiplier for more productive treated firms, g_H, while keeping that of less productive treated firms fixed at $g_L = 1.1$, a 10% productivity gain. Control firms experience no productivity change. In panel (a), the firm–market coefficient from unweighted log-growth regressions increasingly diverges from the firm-level coefficient as treatment heterogeneity rises, reflecting the greater weight placed on high-productivity firms. The two coefficients are close when the productivity multiplier is the same for both types of firm, that is, $g_H = g_L$. In contrast, panel (b) shows that midpoint growth rates estimated with midpoint weights yield identical coefficients across aggregation levels, closely matching the implied firm-level average treatment effect.[8]

3.3.3 Jensen inequality

The second simulation examines the role of Jensen's inequality. As shown in Section 3.2.2, if treatment affects the within-firm dispersion of market growth, the Jensen term differs between

[8] Figure E.1 compares OLS and WLS estimates using midpoint growth rates and shows that WLS is necessary to achieve disaggregation invariance.

Figure 4: Disaggregation invariance: Jensen's inequality



(a) Log growth rate – OLS

(b) Midpoint growth rate – WLS

Note. All firms operate in five markets in both periods. The x-axis varies k_T: higher values raise expected sales and reduce the dispersion of market log growth for treated firms. Control firms keep the same sales distribution in both periods. The two groups have the same distribution when $k_T = k_C = 1$. Bars show 95% confidence intervals for mean estimates across simulation repetitions. Appendix E.1 gives the distributional assumptions.

treated and control firms, creating a gap between firm-level and firm–market coefficients.

To isolate this mechanism, we fix the number of markets at five for every firm, eliminating entry and exit and differences in implicit weights arising from the number of markets. We then vary a parameter (k_T) governing the dispersion of market growth among treated firms. Under our functional assumptions, increasing k_T reduces this dispersion while raising expected sales per market. For control firms, we hold the corresponding parameter fixed at $k_C = 1$.

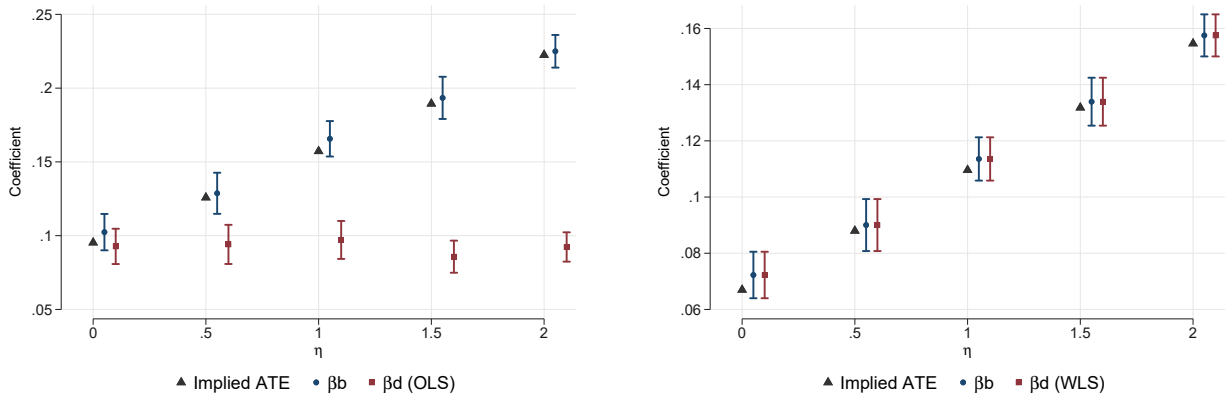
Figure 4 presents our results.[9] As k_T increases, the average treatment effect rises, and both log-growth coefficients in panel (a) increase. However, the decline in the dispersion of market log growth reduces the average Jensen term for treated firms relative to control firms, causing the firm-level coefficient to become progressively smaller than the firm–market coefficient. The two coefficients coincide when treated and control firms have the same distribution of sales in both periods. In contrast, panel (b) shows that midpoint growth rates with midpoint weights continue to deliver identical coefficients across aggregation levels.

3.3.4 Entry and exit

With log growth rates, firm–market regressions are defined only for markets in which the firm is continuously present in both periods. As a result, disaggregated log-growth specifications mechanically exclude entry and exit adjustments. Equations (11) and (12) show how these omitted changes in sales affect the comparison with firm-level growth. We now use the simulation to assess when this source of discrepancy becomes quantitatively important.

[9] Figure E.2 compares log and midpoint growth-rate estimates using WLS.

Figure 5: Disaggregation invariance: entry and exit



(a) Log growth rate – OLS

(b) Midpoint growth rate – WLS

Note. The simulation follows Table E.1. The parameter η, which governs how strongly market participation responds to productivity, varies on the x-axis. Bars show 95% confidence intervals for mean estimates across simulation repetitions.

In our simulation framework, η governs how strongly market participation responds to productivity. When $\eta = 0$, firms still operate in multiple markets, but participation is unaffected by the treatment-induced productivity shock. The extensive margin is therefore present but does not contribute to the treatment effect in expectation. As η increases, treatment induces greater expansion into new markets and reduces exit, so the extensive margin becomes increasingly important in shaping the average treatment effect.

Figure 5 shows the estimation results for different values of η.[10] When $\eta = 0$, the two log-growth coefficients in panel (a) are close, consistent with treatment operating through the intensive margin. As η increases, the gap between firm-level and firm–market coefficients widens, as disaggregated log-growth specifications increasingly miss part of the entry and exit response. Panel (b) shows that the midpoint growth rate always delivers identical firm- and firm–market estimates, which closely match the implied average treatment effect.

When is the extensive margin likely to threaten disaggregation invariance? A simple way to assess the role of the extensive margin in the average treatment effect is to run separate firm-level regressions of the different components of equation (14) on the treatment indicator. Table 1 shows that this methodology delivers an exact decomposition of the average treatment effect into entry, exit, and intensive components.[11] Our findings confirm that the extensive component (entry plus exit) constributes more to the ATE as η increases.

[10] Figure E.3 compares log and midpoint growth-rate estimates using WLS.

[11] See e.g., Beaumont, Libert and Hurlin (2019); Fonseca and Matray (2024); Matray, Müller, Xu and Kabir (2024) for other uses of this decomposition.

Table 1: Decomposition into intensive and extensive margins

η	ATE	Total	Intensive margin	Entry	Exit
0.00	0.07	0.07	0.06	0.00	0.01
		[0.03]	[0.02]	[0.01]	[0.01]
0.50	0.09	0.09	0.06	0.02	0.02
		[0.03]	[0.02]	[0.01]	[0.01]
1.00	0.11	0.11	0.06	0.03	0.02
		[0.03]	[0.03]	[0.01]	[0.01]
1.50	0.13	0.13	0.05	0.05	0.03
		[0.03]	[0.02]	[0.01]	[0.01]
2.00	0.15	0.16	0.06	0.06	0.04
		[0.03]	[0.02]	[0.01]	[0.01]

Note. The simulation follows Table E.1. The parameter η, which governs the response of market participation to productivity, varies across rows. The table reports the implied average treatment effect and firm-level regressions of total midpoint growth, the intensive-margin component, the entry component, and the exit component on treatment.

3.4 Application: detection of confounding shocks

Our simulations show that when disaggregation invariance fails, disaggregated specifications can yield different estimates even in the absence of confounding shocks. In turn, when disaggregation invariance holds, comparisons across levels of aggregation become informative about confounding shocks and their contribution to the baseline average treatment effect. This subsection illustrates how researchers can use disaggregation invariance for this purpose.

We add confounding market shocks to the simulation, with treated firms more likely to operate in markets experiencing negative shocks. Treatment, however, remain as good as randomly assigned conditional on these market shocks. Appendix E.1.3 gives the detailed construction.

A common empirical approach is to compare firm-level estimates with firm–market estimates that include market fixed effects, with larger gaps suggesting more substantial confounding shocks. We compare the ability of standard practice, namely log growth rates estimated by OLS in panel (a), and our approach, namely midpoint growth rates estimated by WLS in panel (b), to detect these confounding shocks.

Table 2 shows that, under the log-growth specification, firm-level and firm–market estimates are very similar even in the presence of confounding shocks. In the simulation, both coefficients are far from the true average treatment effect, yet their similarity may lead the researcher to conclude that market-level shocks are not a concern. This result arises because log-growth specifications fail to capture extensive-margin adjustments. As shown in the previous subsection, disaggregated log regressions omit observations corresponding to entry and exit, leading to a downward bias in the estimated treatment effect. When market shocks are negatively correlated with treatment, firm-level estimates are also biased downward. The two biases are of similar magnitudes, causing

Table 2: Disaggregation invariance: Detecting confounding shocks

Specification	Mean	SE
Panel A: Log growth rate – OLS		
ATE	0.3264	–
$\hat{\beta}^b$	0.1140	0.1720
$\hat{\beta}^d$, no FE	-0.2245	0.2442
$\hat{\beta}^d$, market FE	0.1003	0.0466
Panel B: Midpoint growth rate – WLS		
ATE	0.2267	–
$\hat{\beta}^b$	0.0744	0.1085
$\hat{\beta}^d$, no FE	0.0744	0.1085
$\hat{\beta}^d$, market FE	0.2276	0.0439

Note. The table reports the implied ATE and the mean and standard errors of estimated coefficients across simulations. Treated firms are more exposed to negative market shocks. Firm–market regressions use OLS with log growth rates and WLS with midpoint growth rates. "Market FE" adds market fixed effects. Appendix E.1.3 gives the shock and treatment-assignment construction.

firm-level and firm–market coefficients to appear nearly identical.

By contrast, midpoint growth rates preserve disaggregation invariance and therefore provide a reliable diagnostic for the presence of confounding shocks. The firm–market coefficient increases sharply when market fixed effects are included, correctly indicating that market shocks are negatively correlated with treatment and bringing the estimate close to the implied average treatment effect. An additional advantage of disaggregation invariance is that, because firm-level and firm–market coefficients are aligned, the magnitude of the bias induced by confounding shocks can be directly inferred. In this case, failing to account for market shocks leads to an underestimation of the coefficient by 69%.

3.5 Empirical illustration: local fiscal multiplier

We now illustrate our methodology using real data. We turn to the study of the fiscal multiplier, i.e., how much government spending increases economic activity. We choose this application for two reasons. First, it lends itself naturally to both disaggregation across industries and aggregation across locations. Second, the underlying data are publicly available, allowing us to share the constructed dataset and replication code so that researchers can reproduce the analysis.[12]

Data. We combine U.S. military procurement with county–industry employment and county population. Procurement comes from the Defense Contract Action Data System, available through the National Archives, and from USAspending for the later years. We measure spending using real

[12]Code and data: https://github.com/paulhbeaumont/invariance-replication.

net contract obligations. Employment comes from County Business Patterns, using the publicly available data of Eckert, Fort, Schott and Yang (2021), who impute suppressed employment counts and harmonize industry classifications. These data allow us to observe employment at the county level and within counties up to the six-digit NAICS level. Population comes from the Bureau of Economic Analysis.[13] We harmonize county boundaries and reconcile industry employment with county totals. Our estimation sample contains 3,058 county groups over 1981–2016.

Variable construction. Our outcome is employment growth over three years. Let E_{ct} denote employment in county c in year t. We compare log growth, $\log E_{ct} - \log E_{c,t-3}$, with midpoint growth, $2(E_{ct}-E_{c,t-3})/(E_{ct}+E_{c,t-3})$. County spending is the change in average annual procurement per resident,

$$\text{County spending}_{ct} = \frac{\overline{\text{Spending}}_{ct} - \text{Spending}_{c,t-3}}{\text{Pop}_{c,t-3}},$$

where $\overline{\text{Spending}}_{ct}$ is average annual real procurement over the three years ending in t, and $\text{Pop}_{c,t-3}$ is baseline population. Averaging procurement smooths the uneven timing of contracts, which need not generate spending in every year. We winsorize this spending variable at its first and ninety-ninth percentiles, computed once over all county-years in the sample rather than year by year.

Government spending may itself respond to local economic conditions. To address this reverse-causality concern, we follow the exposure-instrument logic of Nakamura and Steinsson (2014). We interact a county's historical procurement intensity with the national change in spending per resident. Historical intensity is procurement divided by population over a five-year window ending five years before the baseline year. National spending changes use the same three-year average and baseline comparison as county spending. The instrument therefore predicts that a national military-spending expansion will increase procurement more in counties that historically received more military spending. We estimate the employment response by two-stage least squares, including county and year fixed effects throughout. Our methodology naturally extends to the estimation of treatment effects using two-stage least squares, which in this single-instrument setting simply boils down to rescaling the reduced-form coefficient.[14]

Disaggregation and industry shocks. A natural empirical concern is that counties receiving more government spending specialize in industries with different growth prospects. For instance, counties with a larger manufacturing sector may have been more exposed to competition from Chinese imports (Autor, Dorn and Hanson, 2013). To address this concern, we disaggregate em-

[13] The source data are available from the National Archives, USAspending, the Eckert–Fort–Schott–Yang data page, and the BEA county accounts.

[14] Appendix G gives the remaining construction details.

ployment into county–industry cells and include industry–year fixed effects. These effects absorb shocks common to an industry in a given year, while spending and its instrument remain defined at the county level.

Table 3: Investigating the role of industry shocks in the fiscal multiplier

Panel A: Log growth

		County ×			
Level	County	NAICS2	NAICS4	NAICS6	NAICS6
	(1)	(2)	(3)	(4)	(5)
County spending	6.123**	5.104*	5.533*	3.984	2.731
	(2.548)	(2.610)	(3.063)	(2.695)	(1.902)
% deviation	0.0%	-16.7%	-9.6%	-34.9%	-55.4%
County FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Industry × year FE	–	–	–	–	Yes
First-stage F					
County spending	33.41	33.30	32.67	32.46	41.85
Observations	110,088	2,359,648	13,055,068	22,506,999	22,506,999

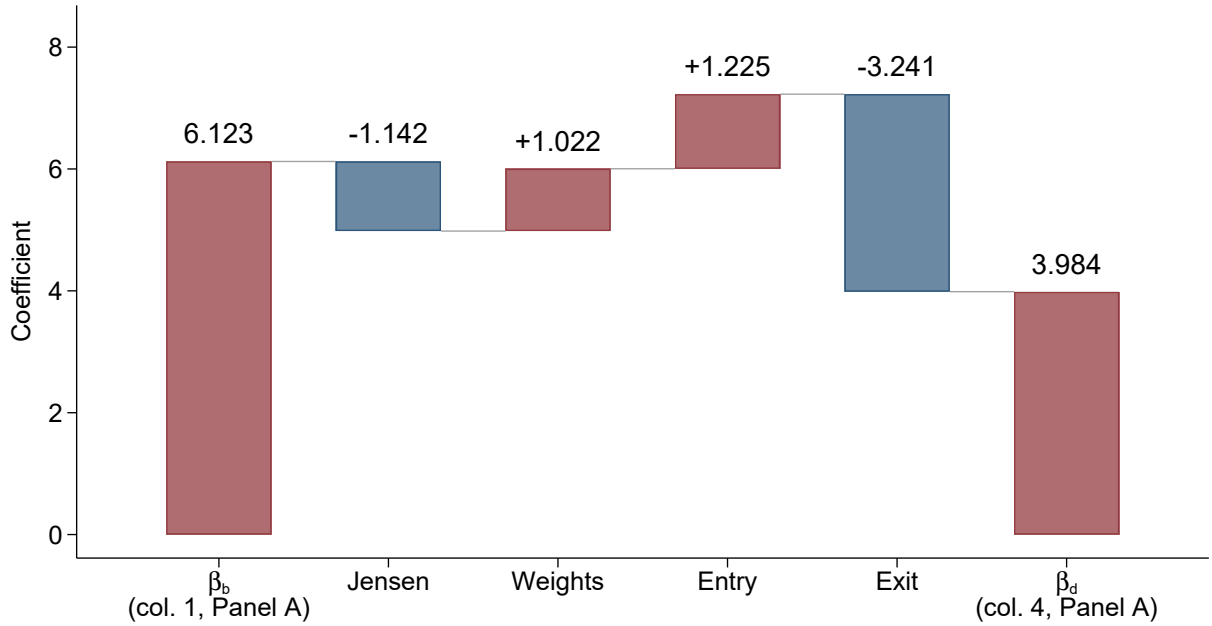
Panel B: Midpoint growth

		County ×			
Level	County	NAICS2	NAICS4	NAICS6	NAICS6
	(1)	(2)	(3)	(4)	(5)
County spending	6.353***	6.353***	6.353***	6.353***	5.388**
	(2.397)	(2.397)	(2.397)	(2.397)	(2.722)
% deviation	0.0%	0.0%	0.0%	0.0%	-15.2%
County FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Industry × year FE	–	–	–	–	Yes
First-stage F					
County spending	33.41	33.41	33.41	33.41	27.41
Observations	110,088	2,551,814	17,664,658	36,517,233	36,517,233

Note. All columns use three-year growth and 2SLS. Coefficients and standard errors are multiplied by 10^5. Parentheses contain county-clustered standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Panel A of Table 3 shows the results using log growth. We weight industry cells by their lagged share of county employment to account for differences in local industry size. As we disaggregate up to six-digit industries (column 4), the coefficient moves away from the county-level value, shrinks by 34.9%, and becomes statistically insignificant even before industry–year fixed effects enter. Therefore, a researcher comparing the county regression with the final specification might conclude that industry shocks account for the county-level relationship between spending and employment growth. In contrast, in Panel B, we show that with midpoint growth and midpoint weights the coefficient remains at 6.35 at every level of disaggregation and falls to 5.39 after including industry–year fixed effects, a decline of about 15%, while remaining statistically significant at the 5% level. Industry shocks therefore only explain away a limited part of the effects of local government spending.

Figure 6: From county to county–industry log-growth coefficients



Note. The graph moves from column (1), β_b, to column (4), β_d, of Panel A in Table 3, before industry–year fixed effects enter. Red bars denote positive changes and blue bars negative changes. Coefficients and components are multiplied by 10^5.

Figure 6 gives the breakdown of the difference between columns (1) and (4) of Panel A. We find that exit contributes the most to the violation of disaggregation invariance, as log growth omits industries in which county employment falls to zero. Relatedly, while lagged employment shares account for local industry size, they cannot recover employment losses in industries omitted from the log-growth regression, with the weights component contributing +1.022 to the difference between the disaggregated and baseline coefficients.

4 Aggregation

4.1 Aggregation invariance

We now turn to aggregation. Researchers often aggregate firm-level outcomes to study whether the gains of treated firms translate into gains for the industry as a whole, or whether they reflect mere reallocation between treated and control firms. For instance, treated firms may expand by taking sales from other firms in the same industry (negative spillovers), reducing the industry response relative to the direct effect on treated firms.

We compare a baseline firm-level regression with industry fixed effects,

$$\Delta S_{it} = \alpha_{j(i)} + \beta^b \, Treated_i + e_{it}, \tag{17}$$

with the industry-level regression

$$\Delta S_{jt} = \alpha^a + \beta^a Share\,Treated_j + e_{jt}, \tag{18}$$

where $j(i)$ denotes firm i's industry. We explain why industry fixed effects are needed in equation (17) in Section 4.2.

Because industries combine firms with potentially different treatment statuses, moving from the firm to the industry level requires defining industry exposure to treatment, $Share\,Treated_j$. A fundamental difference from disaggregation is therefore that we compare regressions with different right-hand-side variables: firm-level treatment at the firm level and exposure to treated firms at the industry level. As we explain below, this change in the regressor requires a different approach to making the estimands comparable.

Firm sales aggregate into industry sales, and treatment exposure is the weighted share of treated firms:

$$S_{jt} = \sum_{i:\, j(i)=j} S_{it}, \qquad Share\,Treated_j = \sum_{i:\, j(i)=j} \omega_{ij} Treated_i, \qquad \sum_{i:\, j(i)=j} \omega_{ij} = 1.$$

We give industries equal weight in the aggregated regression and use the corresponding firm weights in the baseline regression. Each industry therefore receives one unit of weight in total. We discuss the choice of growth rates and weights in more detail in Section 4.3. For the sake of exposition, we start by abstracting from firm entry and exit, and relax this assumption in Section 4.3.

Exogeneity assumptions. Since we use aggregation to detect spillovers, we impose two additional assumptions relative to the disaggregation analysis. First, we assume random assignment of treatment at the firm level, simplifying the interpretation of spillover effects.[15] Second, we assume that industry treatment exposure is unrelated to treatment responses among treated firms.

Both assumptions are important. In particular, the first does not imply the second: even with random assignment at the firm level, industries with greater exposure may contain treated firms that respond differently. Suppose financially constrained firms react less to treatment. If industries with a higher share of treated firms also have a higher share of financially constrained firms among those treated, then $Share\,Treated_j$ will be correlated with industry-level treatment-effect heterogeneity. The industry coefficient can then differ from the firm-level coefficient in the absence of spillovers.

In Section 4.5, we show how researchers can control for observable treatment heterogeneity and

[15] Spillover effects can also be defined and estimated with non-random treatment, but identification then requires additional assumptions about treatment selection and exposure (e.g., Barkley, Hudgens, Clemens, Ali and Emch, 2020). We impose random firm-level treatment to simplify the analysis.

assess the role of unobservable heterogeneity.[16]

Aggregation invariance and SUTVA. Disaggregation decomposes a firm-level outcome into within-firm components, so all disaggregated observations share the same treatment status. Aggregation, by contrast, sums outcomes across firms with potentially different treatment statuses. Industry growth therefore reflects not only the direct effect on treated firms, but also interactions between treated and control firms within the same industry.

The fact that aggregation captures such interactions is precisely why researchers aggregate in the first place: comparing firm- and industry-level estimates is a standard way to detect business stealing, reallocation, or other spillovers within industries. For this comparison to be meaningful, however, the empirical specification must ensure that the firm- and industry-level estimands coincide in the absence of SUTVA violations. Otherwise, differences across aggregation levels may arise mechanically from the change in specification rather than from cross-firm interactions.

These considerations motivate our definition of *aggregation invariance*. Let β^b denote the population coefficient from the firm-level regression with industry fixed effects in equation (17), and β^a the population coefficient from the industry-level regression in equation (18). We say that *aggregation invariance* holds if $\beta^a = \beta^b$ in the absence of SUTVA violations.[17]

4.2 Within- and across-industry variations in treatment exposure

In this subsection, we explain why our definition of aggregation invariance includes industry fixed effects in the firm-level regression (equation (17)). The intuition is that, without industry fixed effects, the baseline firm-level coefficient would also pick up part of the spillover. If, for instance, business stealing reduces firm growth in industries with a larger share of treated firms, the baseline coefficient without fixed effects will understate the direct effect of treatment because it will partly reflect the negative spillovers. Comparing that coefficient with the industry coefficient will then understate the spillover, since part of it is already captured by the baseline regression.

One useful way to see why industry fixed effects are needed is to decompose treatment variation within and across industries. A firm's treatment status can be written as

$$Treated_i = (Treated_i - Share\,Treated_j) + Share\,Treated_j.$$

The first term compares a firm's treatment status with that of other firms in the same industry

[16] In principle, aggregation invariance can be defined without imposing exogeneity. It can be used in accounting exercises to relate firm-level contributions to aggregate outcomes, as in Beaumont, Libert and Hurlin (2019). In this paper, we use the coefficient comparison to detect spillovers, which motivates random treatment assignment and the restriction on treatment heterogeneity.

[17] As with disaggregation, we compare population coefficients: differences across aggregation levels can persist as the sample size grows arbitrarily large.

and, by construction, has weighted mean zero within each industry. By contrast, the second term captures differences in treatment exposure across industries. Industry fixed effects absorb this second source of variation, leaving the first to identify the coefficient on firm-level treatment.

This decomposition suggests an alternative way of identifying spillovers:

$$\Delta S_{it} = \alpha + \beta^b \, Treated_i + \delta Share\,Treated_j + \varepsilon_{it}. \tag{19}$$

Since β^b is identified entirely from within-industry treatment variation, it is the same as the coefficient in the regression with industry fixed effects (equation (17)). The two firm specifications are therefore equivalent for the coefficient comparison. Appendix D.5 gives the formal argument.

We refer to equation (19) as the *exposure-to-treated approach* (e.g., Crépon, Duflo, Gurgand, Rathelot and Zamora, 2013; Cai and Szeidl, 2024; Lerche, 2025). This approach is an alternative to the *aggregation approach* described in equations (17) and (18). Spillovers are detected directly by including $Share\,Treated_j$ in the firm-level regression. The logic of the approach is that a statistically significant coefficient δ indicates that firms' responses depend on their exposure to treated firms, providing evidence of a SUTVA violation.

Averaging equation (19) within industries gives[18]

$$\beta^a = \beta^b + \delta. \tag{20}$$

The industry coefficient captures the direct effect of treatment, β^b, together with the spillover through exposure to treated firms, δ.

Now let β^{naive} denote the coefficient from regressing firm growth on treatment without including industry fixed effects or treatment exposure. Omitting $Share\,Treated_j$ from equation (19) gives

$$\beta^{naive} = \beta^b + \pi\delta, \qquad \pi = \frac{\mathrm{Var}(Share\,Treated_j)}{\mathrm{Var}_\omega(Treated_i)}. \tag{21}$$

Here, π is the share of across-industry variation in total treatment variation, and lies between zero and one.[19]

Combining equations (20) and (21) yields

$$\beta^a - \beta^{naive} = (1-\pi)\delta. \tag{22}$$

[18] Here, industry growth is the corresponding weighted average of firm growth. Section 4.3 explains how the growth rate and weights deliver this relationship.

[19] Firm-level moments use the matching weights, while industry-level moments give each industry equal weight. The omitted-variable formula gives $\pi = \mathrm{Cov}_\omega(Treated_i, Share\,Treated_j)/\mathrm{Var}_\omega(Treated_i)$. Since the treatment share is the weighted mean of treatment within industry, $E_\omega[Treated_i\, Share\,Treated_j] = E_j[(Share\,Treated_j)^2]$ and $E_\omega[Treated_i] = E_j[Share\,Treated_j]$. Hence the covariance in the numerator equals $\mathrm{Var}_j(Share\,Treated_j)$.

When treatment varies both within and across industries, the simple comparison therefore necessarily understates the magnitude of a nonzero spillover. At the extreme, when most treatment variation occurs across industries (i.e., π is close to one), the naive comparison could lead researchers to erroneously conclude that spillovers are absent.

4.3 Restoring aggregation invariance

We first explain why the methods used to achieve disaggregation invariance, Poisson and midpoint growth rates, do not deliver aggregation invariance. We then show how standard growth rates can achieve aggregation invariance using a two-step procedure to accommodate firm entry.

Poisson. Under the firm-level Poisson specification,

$$E(S_{it} \mid i, t) = \exp(\alpha_i + \mu_t + \beta_b \, Treated_i \times Post_t) \, .$$

For the sake of exposition, we focus on an industry j with two firms, firm 0 (untreated) and firm 1 (treated). Aggregating across firms yields

$$E(S_{jt} \mid j, t) = \exp(\mu_t) \left[\exp(\alpha_0) + \exp(\alpha_1 + \beta_b \, Post_t)\right] .$$

The post-period treatment component is therefore governed not by the unweighted share of treated firms, but by the treated firm's share of predicted baseline sales:

$$\frac{\exp(\alpha_1)}{\exp(\alpha_0) + \exp(\alpha_1)}.$$

This object depends on the firm fixed effects rather than only on treatment status. Moreover, the sum of the two exponential means does not preserve the same log-linear specification in industry exposure. Aggregation therefore does not preserve the Poisson treatment coefficient.

Midpoint growth rate. When using the midpoint growth rate, restoring aggregation invariance would require defining industry treatment exposure using midpoint weights, so that $Share\,Treated_j$ reflects the same weights as those governing the aggregation of outcomes. However, these weights depend on contemporaneous sales, $S_{it} + S_{i,t-h}$, and are therefore affected by treatment. As a result, the regressor $Share\,Treated_j$ becomes a function of the outcome, leading to a form of simultaneity bias. This simultaneity bias does not arise in the disaggregation setting, as both the baseline and disaggregated regressions directly use $Treated_i$ as an independent variable.

Standard growth rates. Standard growth rates provide a way to make the two regressions comparable. Unlike midpoint weights, lagged sales shares do not depend on post-treatment outcomes. Using these shares, industry growth can be written as

$$\Delta^s S_{jt} = \sum_{i:\, j(i)=j} \omega_{ij}^s \Delta^s S_{it}, \qquad \omega_{ij}^s = \frac{S_{i,t-h}}{S_{j,t-h}}. \tag{23}$$

We therefore use lagged sales shares both to construct $ShareTreated_j$ and to estimate the firm-level regression by WLS. With this choice, the industry coefficient satisfies $\beta^a = \beta^b + \delta$. In the absence of spillovers, $\delta = 0$ and the two coefficients coincide, delivering aggregation invariance.

When aggregation invariance holds, the aggregation approach and the exposure-to-treated approach are equivalent ways to detect spillovers. This equivalence does not hold in general: if the growth rate and weights do not deliver aggregation invariance, the two approaches can lead to different conclusions.

Accommodating entry and exit. The preceding comparison leaves out sales by new firms. A firm with zero lagged sales has no standard growth rate, so its contribution is not captured by the incumbent regression. We show how our methodology can account for entry using the identity

$$\Delta^s S_{jt} = \underbrace{\sum_{i \in \mathcal{I}_j} \omega_{ij}^s \Delta^s S_{it}}_{\text{Incumbent contribution}} + \underbrace{\frac{Sales_{jt}^{entry}}{S_{j,t-h}}}_{Entry_{jt}}, \tag{24}$$

where $\mathcal{I}_j$ contains all firms in industry j with positive sales at $t-h$. The second term captures current sales by firms with zero lagged sales, expressed relative to total lagged industry sales.

We recover these two components in two steps. First, we estimate the exposure-to-treated regression on incumbent firms using standard growth and lagged sales weights ω_{ij}^s. We define $ShareTreated_j$ from the same firms and weights. The incumbent contribution to the industry coefficient is then $\beta^b + \delta$.

Second, to recover the contribution of new firms to industry growth, we estimate the auxiliary industry regression

$$Entry_{jt} = \alpha^E + \beta^E ShareTreated_j + u_{jt}. \tag{25}$$

Since the regressions for total industry growth and entry use the same industries, exposure measure, and industry weights, equation (24) implies

$$\beta^a = \beta^b + \beta^E + \delta. \tag{26}$$

This distinction is useful economically. Treatment may draw new firms into an industry while

reducing sales at existing firms, so that the two contributions offset in total industry growth. The entry regression separates these changes.

Proposition 2 (Aggregation invariance)**.** *With standard growth rates and lagged sales weights, the two-step procedure described above achieves aggregation invariance. Specifically:*

1. *The exposure-to-treated regression on incumbent firms identifies* $\beta^b + \delta$.
2. *The industry regression for firm entry identifies* β^E.
3. *The industry and firm-level coefficients are linked by* $\beta^a = \beta^b + \beta^E + \delta$*, where* δ *is the additional component associated with spillovers.*

In the absence of spillovers, $\delta = 0$ *and* $\beta^a = \beta^b + \beta^E$.

Inference. The equivalence also extends to inference. In Appendix D.8, we show that inference on $\widehat{\beta}^b + \widehat{\delta} + \widehat{\beta}^E$ matches inference on the industry coefficient when standard errors are clustered by industry.

4.4 When does aggregation invariance matter?

As with disaggregation, implicit weights, nonlinear growth rates, and the extensive margin can cause aggregation invariance to fail. Following the same logic as Section 3.4, we compare the ability of common practices and our approach to detect spillovers. Specifically, we compare the coefficients obtained from the firm-level regression without industry fixed effects or treatment exposure with those from the industry regression in equation (18), using log growth rates estimated by OLS. We first focus on entry and exit, then briefly summarize the results on implicit weights and Jensen's inequality, with detailed simulations in Appendix F.

Entry and exit. Our approach allows the industry-level coefficient to be decomposed into three components: a direct treatment effect, an indirect treatment effect capturing spillovers across firms, and the contribution of firm entry. Without such a decomposition, comparing treatment effect estimates across aggregation levels can be misleading. Similar coefficients at the firm and industry levels do not necessarily imply that spillovers are absent, while differences across levels may reflect aggregation issues rather than indirect effects on incumbent firms.

We illustrate this issue through a simple simulation in which the positive effect of treatment on firm entry partly offsets negative spillovers on incumbent firms. One example is a policy that reduces barriers to entry, encouraging the creation of new firms while increasing competitive pressure on incumbents.

Specifically, we assume that industries with a larger share of treated firms experience more entry but lower sales growth among both treated and control incumbents. Appendix F.4 gives the simulation details.

Table 4: Aggregation invariance: Detecting spillovers

Specification	Mean	SE
Panel A: Log growth rate – OLS		
$\hat{\beta}^a$	0.0268	0.0088
$\hat{\beta}^{naive}$	0.0299	0.0051
Panel B: Standard growth rate – WLS		
$\hat{\beta}^a$	0.0265	0.0090
$\hat{\beta}^b + \hat{\delta} + \hat{\beta}^E$	0.0265	0.0090
$\hat{\delta}$	-0.2405	0.0089
$\hat{\beta}^b$	0.0867	0.0041
$\hat{\beta}^E$	0.1803	0.0027

Note. Appendix F.4 gives the simulation design. The table reports means and standard errors across simulations. Panel B reports the standard-growth decomposition into the industry effect β^a, the spillover effect δ, the direct effect β^b, and the entry effect β^E.

Table 4 reports the simulation results. In Panel A, the firm- and industry-level coefficients are very similar, suggesting little evidence of spillovers. Panel B instead applies the two-step approach presented in Section 4.3. The coefficient on $ShareTreated_j$ in the intensive-margin regression is strongly negative, reflecting the negative spillover imposed on incumbent firms. At the same time, the coefficient on entry is positive, indicating that industries with a larger share of treated firms experience greater firm entry. The positive entry contribution offsets much of the negative incumbent spillover in total industry sales growth.

These results highlight a broader limitation of log-growth specifications for aggregation exercises. Firm-level log regressions exclude entry by construction, whereas the industry outcome includes it. When treatment simultaneously affects entry and incumbent outcomes, comparing firm- and industry-level log coefficients may therefore provide a misleading picture of the underlying economic mechanisms. In contrast, our decomposition based on standard growth rates separates the incumbent spillover from the contribution of entry, making it possible to recover the full aggregate effect.

Implicit weights. As with disaggregation, implicit weights can be a problem when treatment responses differ across industries and industries contain different numbers of firms. Aggregation introduces an additional source of discrepancy, however, because industry growth places greater weight on larger firms, whereas an unweighted firm regression treats all firms equally. This difference becomes important when industries are concentrated (Gabaix, 2011): a shock to one large firm can

substantially move industry sales while having little effect on the average firm's growth. As a result, even with homogeneous treatment effects, firm-specific shocks to large firms can generate substantial differences between firm- and industry-level estimates in finite samples. Our simulation varies concentration through the dispersion of initial firm productivity, holding treatment effects homogeneous. In concentrated industries, the firm-level combination $\beta^b + \delta$ and the industry coefficient β^a can differ substantially under log-growth OLS, with either sign depending on the shocks to large firms. Standard growth with matched WLS weights eliminates the discrepancy in each simulation.

Jensen's inequality. With log growth, industry growth differs from the sales-weighted average of firm growth. In the disaggregation setting, the Jensen term reflects the dispersion of growth across a firm's markets. With aggregation, differences in treatment status within industries provide another reason for aggregation invariance to break through Jensen's inequality. Industries containing both treated and untreated firms will tend to have more dispersed firm growth than industries in which nearly all firms have the same treatment status. The Jensen term therefore depends systematically on the share of treated firms. This mechanism produces larger discrepancies between the log-growth coefficients when treatment shares vary little across industries and remain close to zero or one. In contrast, standard growth with matched weights preserves the coefficient relationship.

4.5 Controlling for treatment heterogeneity

We now relax the restriction that treatment responses among treated firms are unrelated to industry exposure. Even when treatment is random at the firm level, industries with high exposure can contain firms with systematically different treatment responses, causing the firm- and industry-level coefficients to differ in the absence of spillovers. To explain how researchers can account for this heterogeneity, we focus on the exposure-to-treated approach and abstract from entry.

Allowing for treatment heterogeneity, the firm-level exposure-to-treated equation becomes

$$\Delta^s S_{it} = \alpha + \beta_i Treated_i + \delta^{spill} Share\,Treated_j + u_{it}, \tag{27}$$

where β_i is the firm's direct treatment effect and δ^{spill} the spillover. Under firm-level random treatment, the common-effect regression's exposure coefficient δ equals δ^{spill} if

$$\mathrm{Cov}_\omega(\beta_i, Share\,Treated_j \mid Treated_i = 1) = 0. \tag{28}$$

This condition assumes that industries with higher exposure do not systematically contain treated firms with different direct responses. In the following, we outline a three-step procedure to control

for plausible sources of heterogeneity and assess the role of unobservable sources of treatment heterogeneity.

1. Control for known sources of heterogeneity. First, researchers can account for observable, known sources of treatment heterogeneity to neutralize their role when aggregating. Let C_i denote a pre-treatment characteristic that researchers suspect predicts the response to treatment. Controlling for this heterogeneity while preserving aggregation invariance requires including $Treated_i \times C_i$ and its sales-weighted industry average,

$$Share\,TC_j = \sum_{i \in \mathcal{I}_j} \omega_{ij}^s Treated_i C_i, \tag{29}$$

in the firm-level exposure regression. The interaction allows the direct effect to vary with C_i, while the industry average accounts for its distribution across industries. Appendix D.7 shows how to control for treatment heterogeneity in the aggregation approach.

2. Control for size. Researchers do not always know exactly the sources of treatment heterogeneity. In the aggregation setting, a natural starting point is to control for firm size, since it determines firms' contributions to both industry growth and treatment exposure. Since $Share\,Treated_j$ is weighted by sales, an exposure of 20% could mean that one firm accounting for 20% of industry sales is treated, or that many small firms accounting for the same total share are treated. If large firms respond more strongly and account for a larger share of treated activity in higher-exposure industries, the exposure coefficient will be biased upward, incorrectly suggesting positive spillovers.

A simple way to control for size is to divide firms into pre-treatment quintiles using common cutoffs across industries. As in the first step, the regression should include treatment interactions with the size groups and their sales-weighted industry averages to preserve aggregation invariance. Size-group indicators and their industry averages account for differences in baseline growth.

3. Assess covariate balance. How plausible is it that treatment heterogeneity persists after steps 1 and 2? Researchers can examine whether other characteristics of treated firms still vary systematically with exposure. For a pre-treatment characteristic z_i, this involves estimating

$$Treated_i z_i = a + \lambda Treated_i + \pi_z Share\,Treated_j + \sum_{k=1}^{K-1} \rho_k Treated_i B_{ik} + X_{ij}'\kappa + v_{ij}, \tag{30}$$

using the same weights and controls as the outcome regression. Here, B_{ik} indicates firm i's size group. The vector X_{ij} contains the other baseline controls, treatment interactions, and industry averages from steps 1 and 2. The coefficient π_z measures whether industries with higher exposure systematically treat firms with different values of z_i, after accounting for size and the other controls.

In the same way that covariate-balance tests alone cannot establish random treatment, the absence of imbalance in observable characteristics does not rule out a role for unobserved treatment heterogeneity. However, as when discussing the randomness of treatment, showing that industries do not systematically differ in the composition of treated firms does make an explanation based on large unobserved differences less plausible.

4.6 Aggregate spillovers of local fiscal multipliers

As with disaggregation, we apply our methodology to the study of local fiscal multipliers. Local variation in government spending has been used in the literature precisely to assess the aggregate effects of government spending (e.g., Nakamura and Steinsson, 2014; Chodorow-Reich, 2019). We aggregate employment from the same counties to states and compare the county and state responses to spending. Here, the units are counties with positive employment throughout the sample, so county entry and exit are absent by construction.

As recommended above, we use the standard growth rate to measure employment growth and weight county regressions using lagged employment shares. Regional exposure to county spending is the employment-weighted average of the county spending changes defined in Section 3.5. Denoting the county instrument by z_{ct}, regional exposure and its instrument are given by

$$\begin{aligned} \text{Regional exposure}_{st} &= \sum_{c \in s} \omega_{cst} \text{County spending}_{ct}, \\ Z_{st} = \sum_{c \in s} \omega_{cst} z_{ct}, \qquad \omega_{cst} &= \frac{E_{c,t-3}}{E_{s,t-3}}, \end{aligned} \tag{31}$$

where $E_{s,t-3}$ is total employment in state s.

Writing g_{ct} and g_{st} for three-year standard employment growth at the county and state levels, we estimate

$$g_{ct} = \alpha_c + \lambda_t + \beta_b \text{County spending}_{ct} + \delta \text{Regional exposure}_{st} + u_{ct}, \tag{32}$$

$$g_{st} = a_s + \kappa_t + \beta_a \text{Regional exposure}_{st} + v_{st}. \tag{33}$$

The county regression is estimated by 2SLS with weights ω_{cst}, using z_{ct} and Z_{st} as instruments, and it includes county and year fixed effects. The aggregate regression is estimated by equally weighted 2SLS, using Z_{st} as the instrument and including state and year fixed effects. Standard errors are clustered by state.

The estimates in Table 5 indicate that a \$1,000 increase in annual spending per baseline resident corresponds to a 2.66 percentage-point increase in three-year employment growth at the county level, but a 10.83 percentage-point increase at the state level. The difference between the local and

aggregate responses is close to the exposure-to-treated coefficient of 8.08, which corresponds to our spillover estimate.[20] In contrast, a naive comparison of the aggregate regression and the baseline regression omitting regional exposure would suggest spillovers of only 5.45 percentage points per $1,000 increase in government spending per baseline resident.

Table 5: Aggregation of county employment growth

	Baseline β_b (1)	Aggregate β_a (2)	Spillover δ (3)	Naive gap δ^{naive} (4)
	2.664***	10.830***	8.076**	5.446**
	(0.981)	(3.063)	(3.263)	(2.566)
First-stage F				
County spending	41.44	–	41.44	–
Regional exposure	41.86	79.58	41.86	–
Observations	110,088	1,836	110,088	–
Clusters	51	51	51	51

Note. The outcome is three-year standard employment growth. Baseline and Spillover come from the same augmented county 2SLS regression. Aggregate is estimated separately by equally weighted state-level 2SLS. The naive gap subtracts the county coefficient without regional exposure from the state coefficient. County weights are lagged employment shares within states. There are 51 state groups. Standard errors are clustered by state. Coefficients and standard errors are expressed in percentage points per $1,000 (raw coefficients multiplied by 10^5). $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Columns (1) and (2) of Table 6 show that removing county regression weights or using log growth gives slightly lower exposure coefficients, 7.56 and 7.68. This suggests a limited role for implicit weights and Jensen's inequality in this setting. The remaining columns show similar positive spillovers at two- and four-year horizons and when aggregating to component economic areas (CEAs).[21]

In Appendix G.2, we investigate treatment heterogeneity. The balance diagnostic in Figure G.1 suggests that states more exposed to county spending tend to comprise counties with higher employment and income per resident, and lower manufacturing employment shares. We control for these characteristics one at a time in Table G.1.[22] We find similar estimates for the spillover effects, although allowing responses to vary with income substantially reduces precision.

[20] With time-varying employment shares, the weighted average of county fixed effects, $\sum_{c \in s} \omega_{cst} \alpha_c$, varies over time and is not generally absorbed by state fixed effects. Aggregation invariance therefore need not hold exactly in this panel specification. In practice, $\widehat{\beta}_a - \widehat{\beta}_b - \widehat{\delta} = 0.091$, only 0.84% of the state coefficient.

[21] CEAs are smaller geographic groupings. Our sample contains 343 CEAs, compared with 51 state groups.

[22] Each characteristic adds two endogenous interaction terms and two excluded instruments. Controlling for all three characteristics jointly would require eight excluded instruments, compared with two without these controls.

Table 6: Alternative aggregation specifications

	Spillover (δ)				
	(1)	(2)	(3)	(4)	(5)
	7.559*	7.680**	8.870**	7.763*	6.271**
	(4.061)	(3.001)	(3.485)	(4.334)	(2.610)
Growth rate	Standard	Log		Standard	
Horizon h	3	3	4	3	2
Aggregation	State	State	State	CEA	State
County weights	No	Yes	Yes	Yes	Yes
First-stage F					
County spending	24.03	41.44	43.99	11.48	50.78
Regional exposure	25.15	41.86	41.55	3.43	56.80
Observations	110,088	110,088	107,030	110,088	113,146
Clusters	51	51	51	343	51

Note. Each column reports the regional-exposure coefficient from a county 2SLS regression including county spending and county and year fixed effects. Relative to Table 5, columns (1) and (2) change the regression weights and growth measure, respectively. Columns (3)–(5) change the horizon or geography. County weights indicate whether the baseline regression uses weights or not. Regional exposure and its instrument use employment shares throughout. Standard errors are clustered by state except in column (4), where they are clustered by component economic area (CEA). Coefficients and standard errors are multiplied by 10^5. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

5 Conclusion

To what extent are treatment effect estimates comparable across levels of aggregation? We show that changing the level of aggregation can change what a regression estimates, even in the absence of confounding shocks or spillovers.

We first study disaggregation and develop two approaches that preserve the baseline estimand: midpoint growth rates with midpoint weights and Poisson estimation. These approaches achieve disaggregation invariance, allowing researchers to assess how shocks at finer levels contribute to the baseline treatment effect.

Aggregation raises the same concerns but also requires aligning the sources of treatment variation. To detect spillovers, the baseline regression should exploit within-group variation, while the aggregate regression exploits across-group variation. Otherwise, the baseline coefficient already captures part of the spillover, leading naive comparisons to understate its magnitude. We show how standard growth rates with lagged weights restore aggregation invariance, with a separate step to account for entry. We also propose methods to address treatment-effect heterogeneity, which can confound spillover estimates when correlated with aggregate treatment exposure.

In summary, disaggregation and aggregation can provide valuable insights into the mechanisms through which policies operate, but only if differences across levels can be meaningfully interpreted.

By providing simple methods to restore comparability, our paper allows researchers to exploit the benefits of changing aggregation levels without mechanically changing the treatment effects they seek to estimate.

References

Amiti, Mary, and David Weinstein. 2018. "How Much Do Idiosyncratic Bank Shocks Affect Investment? Evidence from Matched Bank-Firm Loan Data." *Journal of Political Economy*, 126(2): 525–587.

Angrist, Joshua D, and Jörn-Steffen Pischke. 2010. "The credibility revolution in empirical economics: How better research design is taking the con out of econometrics." *Journal of economic perspectives*, 24(2): 3–30.

Arkolakis, Costas, Arnaud Costinot, and Andrés Rodríguez-Clare. 2012. "New trade models, same old gains?" *American Economic Review*, 102(1): 94–130.

Auerbach, Alan J., and Yuriy Gorodnichenko. 2012. "Measuring the Output Responses to Fiscal Policy." *American Economic Journal: Economic Policy*, 4(2): 1–27.

Autor, David, David Dorn, and Gordon Hanson. 2013. "The China Syndrome: Local Labor Market Effects of Import Competition in the United States." *American Economic Review*, 103(6): 2121–68.

Baily, Martin Neil, Charles Hulten, David Campbell, Timothy Bresnahan, and Richard E Caves. 1992. "Productivity dynamics in manufacturing plants." *Brookings papers on economic activity. Microeconomics*, 1992: 187–267.

Baker, Andrew C, David F Larcker, and Charles CY Wang. 2022. "How much should we trust staggered difference-in-differences estimates?" *Journal of Financial Economics*, 144(2): 370–395.

Barkley, Brian G., Michael G. Hudgens, John D. Clemens, Mohammad Ali, and Michael E. Emch. 2020. "Causal inference from observational studies with clustered interference, with application to a cholera vaccine study." *The Annals of Applied Statistics*, 14(3): 1432–1448.

Beaumont, Paul, Thibault Libert, and Christophe Hurlin. 2019. "Granular Borrowers." *Working Paper*.

Beck, Thorsten, Samuel Da-Rocha-Lopes, and André F Silva. 2021. "Sharing the pain? Credit supply and real effects of bank bail-ins." *The Review of Financial Studies*, 34(4): 1747–1788.

Berg, Tobias, Markus Reisinger, and Daniel Streitz. 2021. "Spillover effects in empirical corporate finance." *Journal of Financial Economics*, 142(3): 1109–1127.

Berman, Nicolas, Philippe Martin, and Thierry Mayer. 2012. "How do different exporters react to exchange rate changes?" *The Quarterly Journal of Economics*, 127(1): 437–492.

Bertrand, Marianne, Esther Duflo, and Sendhil Mullainathan. 2004. "How much should we trust differences-in-differences estimates?" *The Quarterly Journal of Economics*, 119(1): 249–275.

Blanchard, Olivier, and Roberto Perotti. 2002. "An Empirical Characterization of the Dynamic Effects of Changes in Government Spending and Taxes on Output." *The Quarterly Journal of Economics*, 117(4): 1329–1368.

Borusyak, Kirill, and Peter Hull. 2023. "Nonrandom Exposure to Exogenous Shocks." *Econometrica*, 91(6): 2155–2185.

Borusyak, Kirill, Peter Hull, and Xavier Jaravel. 2021. "Quasi-Experimental Shift-Share Research Designs." *Review of Economic Studies*, 89(1): 181–213.

Boualam, Yasser, and Clément Mazet-Sonilhac. 2025. "The Flow Approach to Credit Markets: Methodology, Measurements, and Macro Perspectives." Working paper.

Cai, Jing, and Adam Szeidl. 2024. "Indirect effects of access to finance." *American Economic Review*, 114(8): 2308–2351.

Callaway, Brantly, and Pedro Sant'Anna. 2021. "Difference-in-Differences with multiple time periods." *Journal of Econometrics*, 225(2): 200–230.

Catherine, Sylvain, Thomas Chaney, Zongbo Huang, David Sraer, and David Thesmar. 2022. "Quantifying reduced-form evidence on collateral constraints." *The Journal of Finance*, 77(4): 2143–2181.

Chemla, Gilles, and Chris Hennessy. 2026. "Discretionary Choice, Information Production, and the SUTVA." *Working paper*.

Chen, Jiafeng, and Jonathan Roth. 2024. "Logs with zeros? Some problems and solutions." *Quarterly Journal of Economics*, 139(2): 891–936.

Chodorow-Reich, Gabriel. 2014. "The employment effects of credit market disruptions: Firm-level evidence from the 2008–9 financial crisis." *The Quarterly Journal of Economics*, 129(1): 1–59.

Chodorow-Reich, Gabriel. 2019. "Geographic cross-sectional fiscal spending multipliers: What have we learned?" *American Economic Journal: Economic Policy*, 11(2): 1–34.

Cingano, Federico, Francesco Manaresi, and Enrico Sette. 2016. "Does credit crunch investment down? New evidence on the real effects of the bank-lending channel." *The Review of Financial Studies*, 29(10): 2737–2773.

Cohn, Jonathan, Zack Liu, and Malcolm Wardlaw. 2022. "Count (and count-like) data in finance." *Journal of Financial Economics*, 146(2): 529–551.

Crépon, Bruno, Esther Duflo, Marc Gurgand, Roland Rathelot, and Philippe Zamora. 2013. "Do labor market policies have displacement effects? Evidence from a clustered randomized experiment." *The quarterly journal of economics*, 128(2): 531–580.

Davis, Steven, and John Haltiwanger. 1992. "Gross Job Creation, Gross Job Destruction, and Employment Reallocation." *Quarterly Journal of Economics*, 107(3): 819–863.

Degryse, Hans, Olivier De Jonghe, Sanja Jakovljević, Klaas Mulier, and Glenn Schepens. 2019. "Identifying Credit Supply Shocks with Bank-Firm Data: Methods and Applications." *Journal of Financial Intermediation*, 40: 100813.

Di Giovanni, Julian, Andrei A Levchenko, and Isabelle Mejean. 2014. "Firms, destinations, and aggregate fluctuations." *Econometrica*, 82(4): 1303–1340.

Di Giovanni, Julian, Andrei A Levchenko, and Isabelle Mejean. 2017. "Large firms and international business cycle comovement." *American Economic Review*, 107(5): 598–602.

Eaton, Jonathan, Samuel Kortum, and Francis Kramarz. 2011. "An Anatomy of International Trade: Evidence from French Firms." *Econometrica*, 79(5): 1453–1498.

Eckert, Fabian, Teresa Fort, Peter Schott, and Natalie Yang. 2021. "Imputing Missing Values in the US Census Bureau's County Business Patterns." National Bureau of Economic Research Working Paper 26632.

Ferracci, Marc, Grégory Jolivet, and Gerard J van den Berg. 2014. "Evidence of treatment spillovers within markets." *Review of Economics and Statistics*, 96(5): 812–823.

Fitzgerald, Doireann, and Stefanie Haller. 2018. "Exporters and shocks." *Journal of International Economics*, 113: 154–171.

Fonseca, Julia, and Adrien Matray. 2024. "Financial Inclusion, Economic Development, and Inequality: Evidence from Brazil." *Journal of Financial Economics*, 156: 1–22.

Foster, Lucia, John C Haltiwanger, and Cornell John Krizan. 2001. "Aggregate productivity growth: Lessons from microeconomic evidence." In *New developments in productivity analysis*. 303–372. University of Chicago Press.

Gabaix, Xavier. 2011. "The granular origins of aggregate fluctuations." *Econometrica*, 79(3): 733–772.

Garthwaite, Craig L. 2014. "Demand spillovers, combative advertising, and celebrity endorsements." *American Economic Journal: Applied Economics*, 6(2): 76–104.

Giroud, Xavier, and Holger Mueller. 2019. "Firms' Internal Networks and Local Economic Shocks." *American Economic Review*, 109(10): 3617–3649.

Goldman, Jim. 2020. "Government as Customer of Last Resort: The Stabilizing Effects of Government Purchases on Firms." *The Review of Financial Studies*, 33(2): 610–643.

Goldsmith-Pinkham, Paul, Isaac Sorkin, and Henry Swift. 2020. "Bartik Instruments: What, When, Why, and How." *American Economic Review*, 110(8): 2586–2624.

Goodman-Bacon, Andrew. 2021. "Difference-in-differences with variation in treatment timing." *Journal of Econometrics*, 225(2): 254–277.

Gormley, Todd, and David Matsa. 2013. "Common Errors: How to (and Not to) Control for Unobserved Heterogeneity." *Review of Financial Studies*, 27: 617–661.

Guren, Adam, Alisdair McKay, Emi Nakamura, and Jón Steinsson. 2021. "What do we learn from cross-regional empirical estimates in macroeconomics?" *NBER Macroeconomics Annual*, 35(1): 175–223.

Helm, Ines. 2020. "National industry trade shocks, local labour markets, and agglomeration spillovers." *The Review of Economic Studies*, 87(3): 1399–1431.

Herreno, Juan. 2023. "Aggregating the Effect of Bank Credit Supply Shocks on Firms."

Hsieh, Chang-Tai, and Peter J Klenow. 2009. "Misallocation and manufacturing TFP in China and India." *The Quarterly journal of economics*, 124(4): 1403–1448.

Huber, Kilian. 2018. "Disentangling the effects of a banking crisis: Evidence from German firms and counties." *American Economic Review*, 108(3): 868–898.

Huber, Kilian. 2023. "Estimating general equilibrium spillovers of large-scale shocks." *The Review of Financial Studies*, 36(4): 1548–1584.

Iyer, Rajkamal, José-Luis Peydró, Samuel da Rocha-Lopes, and Antoinette Schoar. 2014. "Interbank liquidity crunch and the firm credit crunch: Evidence from the 2007–2009 crisis." *The Review of Financial Studies*, 27(1): 347–372.

Jiang, Wei. 2017. "Have instrumental variables brought us closer to the truth." *Review of Corporate Finance Studies*, 6(2): 127–140.

Khwaja, Asim, and Atif Mian. 2008. "Tracing the Impact of Bank Liquidity Shocks: Evidence from an Emerging Market." *American Economic Review*, 98(4): 1413–1442.

Kogan, Leonid, Dimitris Papanikolaou, Amit Seru, and Noah Stoffman. 2017. "Technological innovation, resource allocation, and growth." *The Quarterly Journal of Economics*, 132(2): 665–712.

Lerche, Adrian. 2025. "Direct and indirect effects of investment tax incentives." *American Economic Review*, 115(8): 2781–2818.

Manski, Charles F. 1993. "Identification of endogenous social effects: The reflection problem." *The review of economic studies*, 60(3): 531–542.

Matray, Adrien, Karsten Müller, Chenzi Xu, and Poorya Kabir. 2024. "EXIM's Exit: The Real Effects of Trade Financing by Export Credit Agencies." Working Paper 32019.

McKay, Alisdair, and Christian K Wolf. 2023. "What can time-series regressions tell us about policy counterfactuals?" *Econometrica*, 91(5): 1695–1725.

Mian, Atif, and Amir Sufi. 2014. "What Explains the 2007–2009 Drop in Employment?" *Econometrica*, 82(6): 2197–2223.

Mian, Atif, Andres Pablo Sarto, and Amir Sufi. 2023. "Estimating credit multipliers."

Moulton, Brent R. 1990. "An illustration of a pitfall in estimating the effects of aggregate variables on micro units." *The Review of Economics and Statistics*, 334–338.

Nakamura, Emi, and Jón Steinsson. 2014. "Fiscal Stimulus in a Monetary Union: Evidence from US Regions." *American Economic Review*, 104(3): 753–92.

Nakamura, Emi, and Jón Steinsson. 2018. "Identification in macroeconomics." *Journal of Economic Perspectives*, 32(3): 59–86.

Olley, G Steven, and Ariel Pakes. 1996. "The Dynamics of Productivity in the Telecommunications Equipment Industry." *Econometrica*, 64(6): 1263–1297.

Paravisini, Daniel, Veronica Rappoport, Philipp Schnabl, and Daniel Wolfenzon. 2014. "Dissecting the Effect of Credit Supply on Trade: Evidence from Matched Credit-Export Data." *Review of Economic Studies*, 82(1): 333–359.

Petersen, Mitchell A. 2008. "Estimating standard errors in finance panel data sets: Comparing approaches." *The Review of Financial Studies*, 22(1): 435–480.

Pinardon-Touati, Noémie. 2026. "The Crowding Out Effect of Local Government Debt: Micro- and Macro-Estimates." Working paper.

Schwert, Michael. 2018. "Bank capital and lending relationships." *The Journal of Finance*, 73(2): 787–830.

Silva, Santos, and Silvana Tenreyro. 2006. "The log of Gravity." *Review of Economics and Statistics*, 88(November): 641–658.

Sraer, David, and David Thesmar. 2023. "How to use natural experiments to estimate misallocation." *American Economic Review*, 113(4): 906–938.

Wolf, Christian K. 2023. "The missing intercept: A demand equivalence approach." *American Economic Review*, 113(8): 2232–2269.

Online Appendix for
Zooming In, Zooming Out: Reconciling Treatment Effects across Aggregation Levels

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A Data sources and sample construction

This appendix describes the datasets used in Section 2. For each dataset, we define a baseline level and a disaggregated level, with the latter nested within the former. We also report the data source, sample period, and main construction steps used before computing the descriptive statistics. Across datasets, we impose two common restrictions. First, baseline units must be observed in at least two consecutive periods, so that the analysis focuses on continuing units. Second, observations with missing or non-positive values are dropped before aggregation.

Panjiva–Compustat. The Panjiva–Compustat data combine Panjiva vessel-level bills of lading for shipments departing the United States with Compustat firm identifiers. Shipments are linked to Compustat at the parent-firm level using a gvkey crosswalk. We use the 2013–2021 period. HS codes come from an auxiliary product file that maps each shipment's Harmonized System code to HS07, so we do not apply an additional vintage adjustment. We keep shipments for which the shipper country is the United States and shipment weight is positive. We standardize firm names, drop major non-merchant freight forwarders using name-based regular expressions, split shipment weight equally across all HS6 codes listed on a shipment, and aggregate to the gvkey $\times$ destination $\times$ HS6 product $\times$ year level. The baseline level is the firm, and the disaggregated level is firm $\times$ destination $\times$ HS6 product.

Revelio–Compustat. The Revelio–Compustat data combine employment counts from Revelio Labs with Compustat firm identifiers. Firms are linked at the gvkey level, and occupations are coded using six-digit Standard Occupational Classification codes. We use the 2014–2023 period. We keep firm $\times$ occupation $\times$ year observations with positive headcount. The baseline level is the firm, and the disaggregated level is firm $\times$ occupation.

BACI bilateral trade data. The BACI data come from CEPII and report bilateral trade values at the exporter $\times$ importer $\times$ HS6 product $\times$ year level. We use the HS07 version over the 2007–2022 period. Because CEPII publishes BACI in a fixed HS vintage, no additional product-code crosswalk is applied. We drop within-country flows, keep strictly positive trade values, and aggregate to the exporter $\times$ importer $\times$ HS6 product $\times$ year level. The sample includes eleven large advanced exporters: Australia, Denmark, Finland, France, Germany, Japan, New Zealand, Spain, Switzerland, the United Kingdom, and the United States. The baseline level is the exporter–importer pair, and the disaggregated level is exporter–importer pair $\times$ HS6 product.

U.S. County Business Patterns. The U.S. county-level employment data come from the imputation-corrected County Business Patterns panel of Eckert, Fort, Schott and Yang (2021), originally based on data from the U.S. Census Bureau. Industries are coded using the 2012 North American Industry Classification System. We use county × NAICS6 industry × year observations from 2003 to 2016. We keep observations with positive employment, drop NAICS codes that are not exactly six digits or that contain delimiter characters, and exclude two-digit sectors 11, 92, and 99, corresponding to Agriculture, Forestry, Fishing, and Hunting; Public Administration; and Unclassified industries. The baseline level is the county, and the disaggregated level is county × NAICS6 industry.

13F institutional holdings. The institutional holdings data come from Thomson Reuters S34 13F filings and are linked to monthly stock prices from CRSP using eight-digit CUSIP identifiers. The raw unit is fund manager × stock × quarter. We use the 2000–2024 period. We standardize CUSIP identifiers to eight characters, restrict the sample to common stocks with CRSP share codes 10 or 11, and drop positions below 500,000 U.S. dollars at the reporting date. To address identifier reuse, we split a manager identifier into separate manager identities whenever the manager is inactive for more than four consecutive quarters. The baseline level is the fund manager, and the disaggregated level is fund manager × stock.

Dealscan syndicated loans. The Dealscan syndicated-loan data come from WRDS Dealscan and report syndicated-loan facilities. We follow the lead-bank aggregation conventions in Schwert (2018). We use the 2000–2018 period. The activity measure is new-deal credit, dated to the year in which the deal is signed. Each facility's committed amount is split equally across the lead arrangers on the facility, since the WRDS `bankallocation` field covers only a minority of dollar volume in this vintage. We reserve allocation-based weights for a robustness check.

We keep only lead-arranger lender roles and drop non-bank institutional investors, including collateralized loan obligations, mutual funds, hedge funds, prime funds, pension funds, and other non-bank lenders, identified using the Dealscan institution-type field. We restrict the sample to borrowers in the United States and assign each borrower a four-digit SIC code using Dealscan's `primarysiccode` field, which follows the 1987 SIC vintage. For borrowers without a primary SIC code, we supplement the industry assignment using a name-based regular-expression imputation. We then aggregate facilities to the bank × borrower × year level and to the bank × borrower-SIC4 × year level. We keep banks that satisfy the activity threshold in Schwert (2018), namely at least 50 lead-arranger facilities or 5 billion U.S. dollars in lead volume over the sample window, and that are observed in at least two years.

The lead-bank identifier is time-varying. Each Dealscan lender is linked to a Compustat `gvkey` using the lender concordance in Schwert (2018), including the associated date windows. This procedure allows lender identities to follow their `gvkey` over time and absorbs most ownership changes without manual adjustments. Lenders without a match are assigned first to the Dealscan ultimate-parent identifier and then to the Dealscan company identifier. The baseline level is the bank, and the disaggregated level is bank × borrower-SIC4.

U.S. patents linked to Compustat. The firm-level patent data come from PatentsView bulk tables for granted utility patents and are linked to Compustat using the Kogan, Papanikolaou, Seru and Stoffman (2017) correspondence table and the CRSP–Compustat linking table. We use application years 2003–2013 and date patents by application year rather than grant year. The activity measure is a fractional patent count. Each patent is split equally across its distinct four-character IPC subclasses and assigned to the `gvkey` of its assignee. We keep firm × IPC4 × year cells with positive fractional patent counts and drop firms observed in fewer than two years. The baseline level is the firm, and the disaggregated level is firm × IPC4 subclass.

U.S. patents by county. The county-level patent data come from PatentsView bulk tables for granted utility patents. We use application years 2003–2013 and date patents by application year. Counties are identified using disambiguated inventor locations. The activity measure is a fractional patent count. A patent with inventors located in K distinct counties is split equally across those counties and then split equally across its distinct four-character IPC subclasses. We keep county × IPC4 × year cells with positive fractional patent counts and drop counties observed in fewer than two years. The baseline level is the county, and the disaggregated level is county × IPC4 subclass.

B Computation of simulated average treatment effects

This appendix describes how average treatment effects on firm-level sales growth are computed in our simulations.

Law of iterated expectations. Average growth rates are computed by integrating over the multiple layers of randomness in the data-generating process. Let

$$\Delta^q S_i \equiv \Delta^q(S_{i,2}, S_{i,1})$$

denote a firm-level growth transformation, where q indexes the growth-rate method. For example, $q = l$ corresponds to log growth and $q = mp$ corresponds to midpoint growth.

For a given growth-rate transformation q, the expected growth rate is obtained as

$$E[\Delta^q S_i] = E_{\theta_{i,1}}\Big[E_{K_1,K_2|\theta_{i,1}}\Big[E_{S_1,S_2|K_1,K_2,\theta_{i,1}}[\Delta^q(S_2, S_1)]\Big]\Big].$$

This decomposition highlights three conditioning steps.

(i) *Conditional on initial productivity.* We first fix an initial productivity type

$$\theta_{i,1} = \bar{\theta}_r,$$

which determines both expected market-level sales and expected market participation.

(ii) *Conditional on market participation.* Given productivity, the number of markets served in each period, $(K_{i,1}, K_{i,2})$, is drawn from the Poisson entry process. Conditional on $(K_{i,1}, K_{i,2})$, total sales are sums of $K_{i,t}$ independent exponential draws and therefore follow Gamma distributions.

(iii) *Integrating over sales draws.* Given $(K_{i,1}, K_{i,2}, \theta_{i,1})$, we integrate over the sales draws to compute the expected value of the growth transformation $\Delta^q(S_{i,2}, S_{i,1})$. For log growth, this yields an expression involving the digamma function. For midpoint growth, the expectation can be expressed using a Beta distribution representation and evaluated numerically.

Averaging over productivity. Finally, we integrate over the distribution of initial productivity types. Under the uniform discrete assumption,

$$E[\Delta^q S_i] = \frac{1}{R}\sum_{r=1}^{R} E\left[\Delta^q S_i \mid \theta_{i,1} = \bar{\theta}_r\right].$$

Average treatment effects. The simulated average treatment effect is defined as the difference in expected growth rates between treated and control firms:

$$ATE^q = E[\Delta^q S_i \mid \text{treated}] - E[\Delta^q S_i \mid \text{control}].$$

The two expectations are computed using the same iterated-expectations procedure, but under different period-2 productivity paths. For treated firms, period-2 productivity is

$$\theta_{i,2}^{T} = \theta_{i,1}(1+g),$$

whereas for control firms it is

$$\theta_{i,2}^{C} = \theta_{i,1}.$$

C Do midpoint weights introduce bias?

To what extent is it appropriate to use midpoint weights, given that they depend on contemporaneous sales? Under firm-level random assignment, midpoint weights are completely innocuous: the disaggregated regression targets the same estimand as the baseline firm-level regression, and the latter has a causal interpretation by assumption. The fact that markets in which treated firms expand more receive greater weight is not a distortion introduced by disaggregation, but rather a feature of firm-level growth itself. Firm-level regressions mechanically place more weight on components of sales that account for a larger share of the firm's expansion.

What if treatment is not random at the firm level? The relevant case is when, conditional on market fixed effects, treatment is as good as randomly assigned at the firm–market level. Can midpoint weights make the WLS coefficient biased even though the unweighted OLS coefficient is consistent?

First, note that weights do not change the assignment mechanism. Treatment remains as good as randomly assigned conditional on market fixed effects. Weights can, however, affect the comparison made by the regression by linking the influence of an observation to the residual shock associated with that observation. To see this, consider the weighted treated–control comparison within a given market m. Its residual component is

$$\Delta_m^w = \frac{E(w_{im}u_{im} \mid Treated_i = 1, m)}{E(w_{im} \mid Treated_i = 1, m)} - \frac{E(w_{im}u_{im} \mid Treated_i = 0, m)}{E(w_{im} \mid Treated_i = 0, m)}.$$

The WLS coefficient with market fixed effects can be written as β plus a weighted average of Δ_m^w across markets, with weights determined by the within-market treatment variation and the regression weights.

With OLS, $w_{im} = 1$. Conditional random assignment then implies

$$E(u_{im} \mid Treated_i = 1, m) = E(u_{im} \mid Treated_i = 0, m) = 0,$$

so $\Delta_m^w = 0$ in each market. With midpoint weights, by contrast, the weighted residual terms can be nonzero because the weights may depend on post-treatment sales and therefore on u_{im}. The equation also makes clear, however, that nonzero weighted residuals are not by themselves a problem. Midpoint weights create bias only if the weighted average residual differs between treated and control observations within the same market.

Economically, this condition rules out cases in which midpoint weights assign different importance to idiosyncratic shocks for treated and control firms within the same market. For example, midpoint weights are problematic if treatment leads firms to expand disproportionately in markets where they receive positive residual shocks, because those shocks increase both firm–market growth and the midpoint weight attached to the observation. By contrast, if treated and control firms respond similarly to idiosyncratic shocks within markets, then any relationship between midpoint weights and residual shocks is common across treatment groups and does not bias the treated–control comparison.

Determining whether treated and control firms respond differently to residual shocks cannot be tested directly. In practice, however, a useful diagnostic is to compare midpoint weights w_{im}^{mp} with the weights implied by the standard growth rate, w_{im}^{s}, which are predetermined because they depend only on pre-treatment sales. Since standard-growth weights are defined only for markets in which the firm is present at $t-1$, we compute this comparison on the sample of pre-existing firm–market relationships. If midpoint weights are close to standard weights in this sample, then variation in w_{im}^{mp} primarily reflects predetermined differences in market exposure rather than post-treatment firm–market shocks.

Across the datasets used in Section 2, midpoint weights are highly correlated with standard weights, with an average Pearson correlation of 93.8% (Table C.1 reports the correlation dataset by dataset). While this correlation does not directly test the identifying condition, it suggests that midpoint weights are driven primarily by persistent differences in firm–market size rather than by unexplained firm–market shocks.

Table C.1: Correlation between standard and midpoint weights

Dataset	Baseline	Disaggregated	Average correlation
Panjiva-Compustat	Firm	Firm × HS6 × destination	??
Revelio-Compustat	Firm	Firm × Occupation	??
US patents-Compustat	Firm	Firm × IPC4	??
BACI, destination-level	Exporter × importer	Exporter × importer × HS6	??
United States CBP	County	County × NAICS6	??
US patents-county	County	County × IPC4	??
13F (S34)	Investment manager	Investment manager × Stock	??
Dealscan	Bank	Bank × Borrower SIC4	??

D Proofs

D.1 Decomposition of the log-growth coefficient gap

Consider the baseline and disaggregated log-growth regressions on the same population of firms, each with positive sales in both periods and at least one continuing market. Every regression includes an intercept and the binary treatment indicator $Treated_i$. Both treatment groups have positive probability, and the moments defining the coefficients are finite.

Let $\mathcal{C}_i$ denote firm i's continuing markets and let S^c_{it} and $S^c_{i,t-h}$ be sales summed over those markets in each period. Define

$$w^l_{im} = \frac{S_{im,t-h}}{S^c_{i,t-h}}, \qquad g_{im} = \Delta^l S_{imt}, \qquad n_i = |\mathcal{C}_i|.$$

Thus, the lagged sales shares are normalized over continuing markets, as in Section 3.2.4. Without entry or exit, these are the weights used in Section 3.2.1. As in that subsection, write

$$\bar{g}_i = \frac{1}{n_i} \sum_{m \in \mathcal{C}_i} g_{im}, \qquad a_i = \sum_{m \in \mathcal{C}_i} w^l_{im} g_{im} - \bar{g}_i.$$

Expectations below are over firms.

Implicit weights. With an intercept and a binary treatment indicator, the regression coefficient is the treated–control difference in the relevant group means. Since the sales shares sum to one within each firm, the WLS group mean is

$$\frac{E[\sum_{m \in \mathcal{C}_i} w^l_{im} g_{im} \mid Treated_i = d]}{E[\sum_{m \in \mathcal{C}_i} w^l_{im} \mid Treated_i = d]} = E[\bar{g}_i + a_i \mid Treated_i = d], \qquad d \in \{0, 1\}.$$

The equal-observation OLS group mean is

$$\frac{E[n_i \bar{g}_i \mid Treated_i = d]}{E[n_i \mid Treated_i = d]} = E[\bar{g}_i \mid Treated_i = d] + \frac{\mathrm{Cov}(n_i, \bar{g}_i \mid Treated_i = d)}{E[n_i \mid Treated_i = d]}.$$

Subtracting the OLS mean from the WLS mean within each treatment group, then taking the treated–control difference, proves equation (8).

The within-firm term a_i can also be written as a covariance. Let $\bar{S}_{i,t-h} = S^c_{i,t-h}/n_i$ be average lagged sales per continuing market and define the equal-market covariance using denominator n_i. Then

$$\mathrm{Cov}_{m \in \mathcal{C}_i}(S_{im,t-h}, g_{im}) = \frac{1}{n_i} \sum_{m \in \mathcal{C}_i} S_{im,t-h} g_{im} - \bar{S}_{i,t-h} \bar{g}_i,$$

$$a_i = \frac{\mathrm{Cov}_{m \in \mathcal{C}_i}(S_{im,t-h}, g_{im})}{\bar{S}_{i,t-h}}.$$

Thus, this term captures the relationship between market size and market growth within each firm.

Jensen's inequality. Additivity of continuing-market sales implies

$$\frac{S_{it}^c}{S_{i,t-h}^c} = \sum_{m \in \mathcal{C}_i} w_{im}^l e^{g_{im}}.$$

Taking logs and subtracting weighted average market log growth gives

$$\begin{aligned}\Delta^l S_{it}^c - \sum_{m \in \mathcal{C}_i} w_{im}^l g_{im} &= \log\left(\sum_{m \in \mathcal{C}_i} w_{im}^l e^{g_{im}}\right) - \sum_{m \in \mathcal{C}_i} w_{im}^l g_{im} \\ &= J_{it}^c.\end{aligned}$$

The baseline coefficient $\beta_{b,c}$ is the treated–control difference in average continuing-market log growth. Taking the same difference on both sides therefore gives

$$\beta_{b,c} - \beta_{d,w} = E[J_{it}^c \mid Treated_i = 1] - E[J_{it}^c \mid Treated_i = 0].$$

Entry and exit. By the definitions of the entry and exit shares,

$$S_{it}^c = (1 - e_{it})S_{it}, \qquad S_{i,t-h}^c = (1 - x_{it})S_{i,t-h}.$$

Dividing and taking logs yields

$$\Delta^l S_{it} - \Delta^l S_{it}^c = \log(1 - x_{it}) - \log(1 - e_{it}),$$

which proves equation (11). Taking the treated–control difference gives equation (12).

Combining the contributions. Adding and subtracting the weighted disaggregated coefficient and the baseline coefficient for continuing-market sales gives

$$\beta_b - \beta_{d,OLS} = (\beta_{d,w} - \beta_{d,OLS}) + (\beta_{b,c} - \beta_{d,w}) + (\beta_b - \beta_{b,c}).$$

Substituting the expression for the Jensen contribution proves equation (13).

D.2 Poisson estimator satisfies disaggregation invariance

Disaggregation invariance in Poisson regressions follows from the relationship between the first-order conditions of the disaggregated and firm-level specifications. Let

$$(\widehat{\alpha}_{im}, \widehat{\mu}_t, \widehat{\beta}_d)$$

denote a solution to the first-order conditions of the disaggregated Poisson regression in equation (16). The first-order condition for the treatment effect is

$$\sum_{i,m,t} (Treated_i \times Post_t)(S_{imt} - \widehat{S}_{imt}) = 0,$$

where

$$\widehat{S}_{imt} = \exp(\widehat{\alpha}_{im} + \widehat{\mu}_t + \widehat{\beta}_d \, Treated_i \times Post_t).$$

Because the treatment indicator is common to all markets of a given firm and period, the first-order condition can be aggregated over markets:

$$\sum_{i,t} (Treated_i \times Post_t) \left(\sum_m S_{imt} - \sum_m \widehat{S}_{imt} \right) = 0.$$

Using $S_{it} = \sum_m S_{imt}$, this becomes

$$\sum_{i,t} (Treated_i \times Post_t)(S_{it} - \widehat{S}_{it}) = 0,$$

where

$$\widehat{S}_{it} \equiv \sum_m \widehat{S}_{imt}.$$

We now show that the disaggregated estimates imply a solution to the firm-level first-order conditions. Aggregating the disaggregated fitted values gives

$$\widehat{S}_{it} = \sum_m \exp(\widehat{\alpha}_{im} + \widehat{\mu}_t + \widehat{\beta}_d \, Treated_i \times Post_t).$$

Since $\widehat{\mu}_t$ and $Treated_i \times Post_t$ do not vary across markets within a firm–period, this can be rewritten as

$$\widehat{S}_{it} = \exp(\widehat{\mu}_t + \widehat{\beta}_d \, Treated_i \times Post_t) \sum_m \exp(\widehat{\alpha}_{im}).$$

Equivalently,

$$\widehat{S}_{it} = \exp(\widehat{c}_i + \widehat{\gamma}_t + \widehat{\beta}_b \, Treated_i \times Post_t),$$

where

$$\widehat{c}_i = \log \left(\sum_m \exp(\widehat{\alpha}_{im}) \right), \qquad \widehat{\gamma}_t = \widehat{\mu}_t, \qquad \widehat{\beta}_b = \widehat{\beta}_d.$$

Thus the fitted values constructed from the disaggregated regression have exactly the same exponential form as the fitted values in the baseline firm-level Poisson regression in equation (15).

It remains to verify that these constructed values satisfy the firm-level first-order conditions. For the treatment effect, the previous aggregation implies

$$\sum_{i,t} (Treated_i \times Post_t)(S_{it} - \widehat{S}_{it}) = 0,$$

which is exactly the first-order condition for β_b in the firm-level Poisson regression. The same aggregation argument applies to the fixed-effect first-order conditions. For each firm i,

$$\sum_t (S_{it} - \widehat{S}_{it}) = \sum_t \sum_m (S_{imt} - \widehat{S}_{imt}) = 0,$$

and for each period t,

$$\sum_i (S_{it} - \widehat{S}_{it}) = \sum_i \sum_m (S_{imt} - \widehat{S}_{imt}) = 0.$$

Therefore the constructed coefficient vector

$$(\widehat{c}_i, \widehat{\gamma}_t, \widehat{\beta}_b) = \left(\log \left(\sum_m \exp(\widehat{\alpha}_{im}) \right), \widehat{\mu}_t, \widehat{\beta}_d \right)$$

satisfies the firm-level Poisson first-order conditions.

Under the standard identification conditions for Poisson pseudo-maximum likelihood, the fitted values and the identified treatment coefficient solving the firm-level first-order conditions are unique, up to the usual normalization of fixed effects. Hence the firm-level estimate must satisfy

$$\widehat{\beta}_b = \widehat{\beta}_d.$$

This establishes disaggregation invariance for the Poisson specification.

D.3 Midpoint growth rate with midpoint weights satisfies disaggregation invariance

Consider the baseline WLS regression

$$\Delta^{mp} S_{it} = \alpha^b + \beta^b Treated_i + \epsilon_{it},$$

and the disaggregated WLS regression

$$\Delta^{mp} S_{imt} = \alpha^d + \beta^d Treated_i + \epsilon_{imt}.$$

The disaggregated regression is estimated using midpoint weights. Let

$$z_i = \begin{pmatrix} 1 \\ Treated_i \end{pmatrix}$$

denote the regressor vector. Since treatment is defined at the firm level, z_i is common to all markets m of firm i.

The disaggregated WLS first-order condition is

$$\sum_i \sum_{m \in m(i)} w_{im}^{mp} z_i \left(\Delta^{mp} S_{imt} - z_i' \widehat{\theta}^d \right) = 0,$$

where

$$\widehat{\theta}^d = \begin{pmatrix} \widehat{\alpha}^d \\ \widehat{\beta}^d \end{pmatrix}.$$

Because z_i does not vary across markets within firm, this can be rewritten as

$$\sum_i z_i \left[\sum_{m \in m(i)} w_{im}^{mp} \left(\Delta^{mp} S_{imt} - z_i' \widehat{\theta}^d \right) \right] = 0.$$

Using the normalization of the weights and the midpoint-growth aggregation property,

$$\sum_{m \in m(i)} w_{im}^{mp} \Delta^{mp} S_{imt} = \Delta^{mp} S_{it},$$

we obtain

$$\sum_i z_i \left(\Delta^{mp} S_{it} - z_i' \widehat{\theta}^d \right) = 0.$$

This is exactly the first-order condition of the baseline OLS regression,

$$\sum_i z_i \left(\Delta^{mp} S_{it} - z_i' \widehat{\theta}^b \right) = 0,$$

where

$$\widehat{\theta}^b = \begin{pmatrix} \widehat{\alpha}^b \\ \widehat{\beta}^b \end{pmatrix}.$$

Therefore, the disaggregated estimates satisfy the baseline first-order conditions. Under the usual full-rank condition for the weighted regressor matrix, the OLS solution is unique. Hence,

$$\widehat{\theta}^d = \widehat{\theta}^b.$$

Equivalently,

$$\widehat{\alpha}^d = \widehat{\alpha}^b, \qquad \widehat{\beta}^d = \widehat{\beta}^b.$$

D.4 Inference under disaggregation invariance

We now show that the same aggregation logic implies equality of the cluster-robust variance matrix when clustering is performed at the baseline level. Consider a WLS regression with coefficient vector θ. The cluster-robust variance estimator can be written in sandwich form as

$$\widehat{V}_{cl}(\widehat{\theta}) = \widehat{A}^{-1} \widehat{B} \widehat{A}^{-1},$$

where the bread is

$$\widehat{A} = \sum_\ell w_\ell z_\ell z_\ell',$$

and the meat is

$$\widehat{B} = \sum_c \widehat{g}_c \widehat{g}_c', \qquad \widehat{g}_c = \sum_{\ell \in c} w_\ell z_\ell \widehat{\epsilon}_\ell.$$

Here, ℓ indexes observations, c indexes clusters, z_ℓ is the regressor vector, w_ℓ is the WLS weight, and $\widehat{\epsilon}_\ell$ is the estimated residual.

We compare the sandwich components in the baseline and disaggregated regressions. Because clustering is performed at the firm level, the cluster is i. The firm-level cluster score in the disaggregated regression is

$$\widehat{g}_i^d = \sum_{m \in m(i)} w_{im}^{mp} z_i \widehat{\epsilon}_{imt},$$

where

$$\widehat{\epsilon}_{imt} = \Delta^{mp} S_{imt} - z_i' \widehat{\theta}^d.$$

Since z_i is constant across markets within firm,

$$\begin{aligned}
\widehat{g}_i^d &= z_i \sum_{m \in m(i)} w_{im}^{mp} \left(\Delta^{mp} S_{imt} - z_i' \widehat{\theta}^d \right) \\
&= z_i \left[\sum_{m \in m(i)} w_{im}^{mp} \Delta^{mp} S_{imt} - z_i' \widehat{\theta}^d \sum_{m \in m(i)} w_{im}^{mp} \right].
\end{aligned}$$

Using the midpoint-growth aggregation property and the normalization of the weights, this becomes

$$\widehat{g}_i^d = z_i \left(\Delta^{mp} S_{it} - z_i' \widehat{\theta}^d \right).$$

By the coefficient invariance result above, $\widehat{\theta}^d = \widehat{\theta}^b$. Therefore,

$$\widehat{g}_i^d = z_i \left(\Delta^{mp} S_{it} - z_i' \widehat{\theta}^b \right) = \widehat{g}_i^b,$$

where $\widehat{g}_i^b$ is the firm-level cluster score in the baseline regression. Hence,

$$\widehat{g}_i^d = \widehat{g}_i^b$$

for every firm i. It follows that the meat is identical across the two regressions:

$$\widehat{B}^d = \sum_i \widehat{g}_i^d \widehat{g}_i^{d\prime} = \sum_i \widehat{g}_i^b \widehat{g}_i^{b\prime} = \widehat{B}^b.$$

The bread is also identical. In the disaggregated regression,

$$\widehat{A}^d = \sum_i \sum_{m \in m(i)} w_{im}^{mp} z_i z_i'.$$

Since z_i does not vary across markets within firm and the weights sum to one within firm,

$$\widehat{A}^d = \sum_i z_i z_i' \sum_{m \in m(i)} w_{im}^{mp} = \sum_i z_i z_i' = \widehat{A}^b.$$

Thus,

$$\widehat{A}^d = \widehat{A}^b \qquad \text{and} \qquad \widehat{B}^d = \widehat{B}^b.$$

Therefore,

$$\widehat{V}^d_{cl(i)}(\widehat{\theta}) = \left(\widehat{A}^d\right)^{-1} \widehat{B}^d \left(\widehat{A}^d\right)^{-1} = \left(\widehat{A}^b\right)^{-1} \widehat{B}^b \left(\widehat{A}^b\right)^{-1} = \widehat{V}^b_{cl(i)}(\widehat{\theta}).$$

The equality holds for the full variance-covariance matrix. Therefore, it also holds for the variance of the treatment coefficient:

$$\widehat{V}^d_{cl(i)}(\widehat{\beta}) = \widehat{V}^b_{cl(i)}(\widehat{\beta}),$$

and hence

$$\widehat{SE}^d_{cl(i)}(\widehat{\beta}) = \widehat{SE}^b_{cl(i)}(\widehat{\beta}).$$

D.5 Aggregation invariance

We first consider incumbent firms and then add entry. Define industry treatment exposure by

$$ShareTreated_j = \sum_{i \in \mathcal{I}_j} \omega^s_{ij} Treated_i, \qquad \omega^s_{ij} = \frac{S_{i,t-h}}{S_{j,t-h}}, \qquad \sum_{i \in \mathcal{I}_j} \omega^s_{ij} = 1.$$

Let E_j denote an expectation giving industries equal weight, and define firm-level weighted moments by

$$E_\omega[f_{ij}] = E_j \left[\sum_{i \in \mathcal{I}_j} \omega^s_{ij} f_{ij} \right].$$

The same argument applies with industry weights v_j when firm weights are $v_j \omega^s_{ij}$. Assume the moments below exist and the regressor matrices have full rank. In particular, exposure varies across industries and there is within-industry treatment variation.

Industry fixed effects and industry exposure. Define the within-industry treatment component

$$d_{ij} = Treated_i - ShareTreated_j.$$

It has weighted mean zero in every industry, so $E_\omega[d_{ij} f_j] = 0$ for any industry-level variable f_j. The baseline fixed-effect coefficient is consequently

$$\beta^b_{FE} = \frac{E_\omega[d_{ij} \Delta^s S_{it}]}{E_\omega[d^2_{ij}]}.$$

The exposure-augmented regression can be written as

$$\Delta^s S_{it} = \alpha + \beta^b (Treated_i - ShareTreated_j) + (\beta^b + \delta) ShareTreated_j + \varepsilon_{it}.$$

Because d_{ij} is orthogonal to the constant and $ShareTreated_j$, its coefficient is the same ratio. Thus, $\beta^b_{FE} = \beta^b$. This equality holds for arbitrary outcomes and does not require treatment-effect homogeneity.

The coefficient relationship. Let

$$Y_j^I = \sum_{i \in \mathcal{I}_j} \omega_{ij}^s \Delta^s S_{it}, \qquad x_j = \begin{pmatrix} 1 \\ Share\,Treated_j \end{pmatrix}, \qquad \theta^f = \begin{pmatrix} \alpha \\ \beta^b + \delta \end{pmatrix}.$$

The first-order conditions for the constant and exposure in the reparameterized firm regression are

$$E_j \left[\sum_{i \in \mathcal{I}_j} \omega_{ij}^s x_j (\Delta^s S_{it} - x_j' \theta^f - \beta^b d_{ij}) \right] = 0.$$

The within-industry component drops out, leaving

$$E_j[x_j(Y_j^I - x_j'\theta^f)] = 0.$$

This is the population normal equation for the industry regression of the incumbent contribution Y_j^I on a constant and $Share\,Treated_j$. Uniqueness of the projection implies that its coefficient is $\beta^b + \delta$. Without entry, $Y_j^I = \Delta^s S_{jt}$, proving equation (20).

Entry and exit. Multiplying each incumbent's standard growth by its lagged sales share gives

$$Y_j^I = \frac{\sum_{i \in \mathcal{I}_j} (S_{it} - S_{i,t-h})}{S_{j,t-h}}.$$

The sum includes firms with zero current sales. Adding current sales of new firms gives total industry growth:

$$\Delta^s S_{jt} = Y_j^I + Entry_{jt}.$$

The coefficient from projecting a sum of outcomes on the same regressors equals the sum of their coefficients. Therefore,

$$\beta^a = \beta^b + \delta + \beta^E.$$

This proves equation (26).

D.6 Attenuation in the unadjusted comparison

Use the same incumbent outcomes, weights, and exposure as in Appendix D.5. Let β^{naive} be the coefficient from projecting firm growth on a constant and firm-level treatment alone. The residual from the augmented regression is orthogonal to firm-level treatment, so

$$\beta^{naive} = \beta^b + \delta \frac{\mathrm{Cov}_\omega(Treated_i, Share\,Treated_j)}{\mathrm{Var}_\omega(Treated_i)}.$$

Since industry exposure is the weighted mean of treatment,

$$E_\omega[Treated_i Share\,Treated_j] = E_j[Share\,Treated_j^2], \qquad E_\omega[Treated_i] = E_j[Share\,Treated_j].$$

It follows that

$$\mathrm{Cov}_\omega(Treated_i, Share\,Treated_j) = \mathrm{Var}_j(Share\,Treated_j).$$

The orthogonal decomposition $Treated_i = d_{ij} + Share\,Treated_j$ also yields

$$\mathrm{Var}_\omega(Treated_i) = E_\omega[d_{ij}^2] + \mathrm{Var}_j(Share\,Treated_j).$$

Thus, $\pi = \mathrm{Var}_j(Share\,Treated_j)/\mathrm{Var}_\omega(Treated_i)$ lies in $[0,1]$, and lies strictly inside that interval under the rank conditions stated above. Combining with the coefficient relationship gives

$$\beta^a - \beta^{naive} = (1-\pi)\delta$$

without entry. With entry, the right-hand side becomes $(1-\pi)\delta + \beta^E$. Under the maintained causal interpretation of δ, the no-entry gap has its sign but a smaller absolute magnitude whenever $\delta \neq 0$ and $0 < \pi < 1$.

D.7 Matching controls across levels

Let Z_{ij} be a vector of firm-level controls, including treatment interactions used to allow for heterogeneous direct effects. Define its industry average and within-industry deviation:

$$R_j = \sum_{i \in \mathcal{I}_j} \omega_{ij}^s Z_{ij}, \qquad Z_{ij}^W = Z_{ij} - R_j.$$

For example, when $Z_{ij} = Treated_i C_i$, its industry average is $Share\,TC_j$ from equation (29). Use a linearly independent set of controls. In particular, omit one treatment-by-size category, since the full set sums to $Treated_i$, and its full set of industry averages sums to $Share\,Treated_j$.

Consider the matched exposure-augmented regression

$$\Delta^s S_{it} = \alpha + \beta^b Treated_i + \delta Share\,Treated_j + \gamma' Z_{ij} + \eta' R_j + \varepsilon_{it}. \tag{D.1}$$

Its within–between representation is

$$\Delta^s S_{it} = \alpha + \beta^b(Treated_i - Share\,Treated_j) + \gamma' Z_{ij}^W + (\beta^b + \delta) Share\,Treated_j + (\gamma + \eta)' R_j + \varepsilon_{it}.$$

Both within-industry regressors have weighted mean zero in every industry. Their weighted cross-products with the constant, $Share\,Treated_j$, and R_j are therefore zero. The normal equations separate into a within-industry block and an industry-level block.

The within block is exactly that of the fixed-effect baseline

$$\Delta^s S_{it} = \alpha_j + \beta_{FE}^b Treated_i + \gamma_{FE}' Z_{ij} + e_{it}.$$

Thus, $\beta_{FE}^b = \beta^b$ and $\gamma_{FE} = \gamma$. The industry block is the projection

$$Y_j^I = \alpha^a + \beta_I^a Share\,Treated_j + \lambda' R_j + e_{jt},$$

so that

$$\beta_I^a = \beta^b + \delta, \qquad \lambda = \gamma + \eta.$$

With entry, project both total industry growth and $Entry_{jt}$ on the same constant, $Share\,Treated_j$, and R_j. The coefficient relationship remains

$$\beta^a = \beta^b + \delta + \beta^E.$$

Thus, adding firm controls alone is not enough to preserve the exact comparison. Their industry averages must also enter the exposure-augmented firm regression and the industry regressions.

D.8 Inference under aggregation invariance

This section shows that, under the specification presented in Section 4.3, the standard error around the firm-level linear combination $\widehat{\beta}^b + \widehat{\delta}$ is the same as the standard error around the industry-level coefficient $\widehat{\beta}^a$ when standard errors are clustered at the industry level. We first consider incumbent growth without entry. The equality uses the unadjusted sandwich estimators, or the same finite-sample correction at both levels. Consider the firm-level WLS regression

$$\Delta^s S_{it} = \alpha^b + \beta^b Treated_i + \delta Share\,Treated_j + u_{it}^b,$$

estimated with weights ω_{ij}^s, with standard errors clustered at the industry level. We want inference on

$$\theta \equiv \beta^b + \delta.$$

As in Appendix D.5, we rewrite the regression as

$$\Delta^s S_{it} = \alpha^b + \beta^b \left(Treated_i - Share\,Treated_j\right) + \theta Share\,Treated_j + u_{it}^b.$$

The industry-level regression is

$$\Delta^s S_{jt} = \alpha^a + \beta^a Share\,Treated_j + u_{jt}^a.$$

Equality of the cluster-level scores. Focus on the score contribution for the aggregated block consisting of the constant and the coefficient on $Share\,Treated_j$. In the reparameterized firm-level regression, this coefficient is θ; in the industry-level regression, it is β^a. Let

$$Z_j = \begin{pmatrix} 1 \\ Share\,Treated_j \end{pmatrix}.$$

At the firm level, the industry-clustered score contribution for this block is

$$\widehat{g}_j^b = \sum_{i:j(i)=j} \omega_{ij}^s Z_j \widehat{u}_{it}^b,$$

where

$$\widehat{u}_{it}^{b} = \Delta^{s} S_{it} - \widehat{\alpha}^{b} - \widehat{\beta}^{b} \left(Treated_{i} - Share\, Treated_{j} \right) - \widehat{\theta} Share\, Treated_{j}.$$

Since Z_j is constant within industry, substituting the residual into the score gives

$$\widehat{g}_{j}^{b} = Z_{j} \Bigg[\sum_{i:j(i)=j} \omega_{ij}^{s} \Delta^{s} S_{it} - \widehat{\alpha}^{b} \sum_{i:j(i)=j} \omega_{ij}^{s} - \widehat{\beta}^{b} \sum_{i:j(i)=j} \omega_{ij}^{s} \left(Treated_{i} - Share\, Treated_{j} \right) - \widehat{\theta} Share\, Treated_{j} \sum_{i:j(i)=j} \omega_{ij}^{s} \Bigg].$$

Using the aggregation properties of the standard growth rate,

$$\sum_{i:j(i)=j} \omega_{ij}^{s} \Delta^{s} S_{it} = \Delta^{s} S_{jt}, \qquad \sum_{i:j(i)=j} \omega_{ij}^{s} = 1,$$

and

$$\sum_{i:j(i)=j} \omega_{ij}^{s} \left(Treated_{i} - Share\, Treated_{j} \right) = 0,$$

we obtain

$$\widehat{g}_{j}^{b} = Z_{j} \left(\Delta^{s} S_{jt} - \widehat{\alpha}^{b} - \widehat{\theta} Share\, Treated_{j} \right).$$

Using the fact that the specification delivers aggregation invariance,

$$\widehat{\alpha}^{b} = \widehat{\alpha}^{a}, \qquad \widehat{\theta} = \widehat{\beta}^{a},$$

so

$$\widehat{g}_{j}^{b} = Z_{j} \left(\Delta^{s} S_{jt} - \widehat{\alpha}^{a} - \widehat{\beta}^{a} Share\, Treated_{j} \right).$$

The right-hand side is exactly the industry-level score contribution,

$$\widehat{g}_{j}^{a} = Z_{j} \widehat{u}_{jt}^{a} = Z_{j} \left(\Delta^{s} S_{jt} - \widehat{\alpha}^{a} - \widehat{\beta}^{a} Share\, Treated_{j} \right).$$

Therefore,

$$\widehat{g}_{j}^{b} = \widehat{g}_{j}^{a}$$

for every industry j. It follows that the meat of the industry-clustered sandwich estimator is identical for the aggregated block:

$$\widehat{B}^{b} = \sum_{j} \widehat{g}_{j}^{b} \widehat{g}_{j}^{b\prime} = \sum_{j} \widehat{g}_{j}^{a} \widehat{g}_{j}^{a\prime} = \widehat{B}^{a}.$$

Equality of the bread. The bread is also identical for the aggregated block. In the firm-level WLS regression, the weighted second-moment matrix for the constant and $ShareTreated_j$ is

$$\sum_j \sum_{i:j(i)=j} \omega_{ij}^s Z_j Z_j'.$$

Since Z_j is constant within industry and the weights sum to one within industry,

$$\sum_j \sum_{i:j(i)=j} \omega_{ij}^s Z_j Z_j' = \sum_j Z_j Z_j' \sum_{i:j(i)=j} \omega_{ij}^s = \sum_j Z_j Z_j'.$$

This is exactly the corresponding industry-level bread for the constant and $ShareTreated_j$.

The additional firm-level regressor,

$$Treated_i - ShareTreated_j,$$

does not affect the bread for the aggregated block because it aggregates to zero within industry:

$$\sum_{i:j(i)=j} \omega_{ij}^s \left(Treated_i - ShareTreated_j\right) = 0.$$

Thus, the weighted cross-product between the aggregated block and $Treated_i - ShareTreated_j$ is

$$\begin{aligned}\sum_j \sum_{i:j(i)=j} \omega_{ij}^s Z_j \left(Treated_i - ShareTreated_j\right) &= \sum_j Z_j \sum_{i:j(i)=j} \omega_{ij}^s \left(Treated_i - ShareTreated_j\right) \\ &= 0.\end{aligned}$$

Hence, the bread for the aggregated block is the same at the firm and industry levels:

$$\widehat{A}^b = \widehat{A}^a.$$

Since both the bread and the meat are identical for the aggregated block,

$$\widehat{V}_{cl(j)}^b = \left(\widehat{A}^b\right)^{-1} \widehat{B}^b \left(\widehat{A}^b\right)^{-1} = \left(\widehat{A}^a\right)^{-1} \widehat{B}^a \left(\widehat{A}^a\right)^{-1} = \widehat{V}_{cl(j)}^a.$$

The variance of $\widehat{\theta} = \widehat{\beta}^b + \widehat{\delta}$ is the $(2,2)$ element of this matrix, which is identical to the variance of $\widehat{\beta}^a$. Therefore,

$$\widehat{SE}_{cl(j)}^b \left(\widehat{\beta}^b + \widehat{\delta}\right) = \widehat{SE}_{cl(j)}^a \left(\widehat{\beta}^a\right).$$

Adding entry. Let $x_j = (1, ShareTreated_j)'$, and let $A = \sum_j x_j x_j'$. The industry coefficient vector for total growth is the sum of the vectors from projecting Y_j^I and $Entry_{jt}$ on x_j. Its residual is consequently $\widehat{e}_j^I + \widehat{e}_j^E$. The corresponding score is

$$x_j(\widehat{e}_j^I + \widehat{e}_j^E).$$

The firm-level scores for the reparameterized incumbent block equal $x_j \widehat{e}_j^I$, as shown above. Stacking those scores with the entry regression therefore gives the same variance for the sum as the total-growth regression:

$$A^{-1}\left[\sum_j x_j x_j'(\widehat{e}_j^I + \widehat{e}_j^E)^2\right] A^{-1}.$$

E Additional results

E.1 Disaggregation simulations: design and implementation

This appendix gives the simulation details omitted from Sections 3.3.1–3.4. We first describe the common data-generating process, then the changes used to isolate each source of disaggregation-invariance failure and to introduce confounding shocks.

E.1.1 Common data-generating process

Firm productivity. We consider a collection of firms indexed by i observed in two periods $t \in \{1, 2\}$. Each firm is characterized by productivity θ_{it}. In the pre-treatment period, initial productivity $\theta_{i,1}$ is heterogeneous and drawn uniformly from a finite set of values. To isolate the effect of treatment, we assume that half of firms are treated between periods 1 and 2. Treatment takes the form of a proportional productivity increase of size $g > 0$:

$$\theta_{i,2}^{Treated} = \theta_{i,1}(1 + g), \qquad \theta_{i,2}^{Control} = \theta_{i,1}.$$

Sales per market. Conditional on operating in a market, firm i draws sales in market m at time t from an exponential distribution:

$$E(S_{imt} \mid \theta_{it}) = \mu_0 \, \theta_{it}^{\gamma}.$$

The parameter γ governs the elasticity of sales to productivity and can be interpreted as capturing how strongly productivity translates into market-level revenue. When γ is small, sales are weakly related to productivity. One interpretation of a low γ could be the presence of economic frictions (e.g., switching costs) preventing firm competitive advantages from translating into revenue differences.

Number of markets (extensive margin). Firm productivity also affects the extensive margin. We model the number of active markets as[1]

$$E(K_{it} \mid \theta_{it}) = 1 + \lambda_0 \, \theta_{it}^{\eta}.$$

The parameter η governs how strongly market participation responds to productivity. Larger values of η imply that differences in productivity translate into larger differences in the number of markets in which firms operate, for instance due to the presence of important market fixed costs of entry.

[1] We model the number of active markets as one plus a Poisson distribution to ensure that firms always operate in at least one market. This allows us to abstract from firm-level entry and exit, which would unnecessarily complicate the analysis.

Total firm sales. Firm-level sales aggregate across all markets in which the firm operates. Taking expectations yields

$$E(S_{it} \mid \theta_{it}) = E(K_{it} \mid \theta_{it}) \cdot E(S_{imt} \mid \theta_{it}) = (1 + \lambda_0 \, \theta_{it}^{\eta}) \, \mu_0 \, \theta_{it}^{\gamma}.$$

Thus, treatment increases firm-level sales through two channels: higher sales per market and expansion in the number of markets.

Table E.1: Disaggregation simulations: baseline parameters

Parameter	Description	Default
Initial firm productivity		
θ_1	Initial productivity	1
Firm–market sales: $S \sim \text{Exponential}(\mu_0 \, \theta^{\gamma})$		
μ_0	Baseline sales multiplier	1
γ	Sensitivity to productivity	1
Number of markets per firm: $K \sim \text{Poisson}(\lambda_0 \, \theta^{\eta}) + 1$		
λ_0	Baseline market multiplier	1
η	Sensitivity to productivity	1
Treatment: $\theta_2 = (1 + g) \cdot \theta_1$		
g_{control}	Productivity upgrade for control firms	0
g_{treated}	Productivity upgrade for treated firms	0.1
Simulation specifications		
N_{firm}	Number of firms per simulation	5,000
N_{loop}	Number of simulations	50

Implied average treatment effects. The framework generates sales in levels in both periods, allowing us to evaluate different growth transformations using the same underlying parameters. We compute the implied firm-level average treatment effects under log growth and midpoint growth using the derivations in Appendix B. The average treatment effect is not defined for the standard growth rate under our simulation parameters.

Simulation implementation. For each parameter configuration, we generate simulated datasets 50 times and each time estimate both the baseline firm-level regression and its disaggregated firm–market counterpart. We report the average estimated coefficient across repetitions for each specification. The figures show 95% confidence intervals for these means, computed as the mean plus or minus 1.96 times the standard deviation across repetitions divided by the square root of the number of repetitions. Repeating the simulation makes systematic differences in estimands easier to distinguish from simulation noise.

Unless stated otherwise, the disaggregation simulations use the parameters summarized in Table E.1. Our parameter choices imply that on average in the pre-treatment period, firms are present in 2 markets and sell $1 million per market.

E.1.2 Channel-specific simulations

Implicit weights. Firms draw low or high initial productivity with equal probability, with $\theta_L = 0.1$ and $\theta_H = 3$. Pre-period productivity determines market counts using $\eta = 3$, and these counts are then held fixed across periods. In period 2, treated low-productivity firms receive a productivity multiplier $g_L = 1.1$, while treated high-productivity firms receive the multiplier g_H varied in Figure 3. Thus, g_L and g_H are multipliers, rather than the proportional increase g used in the common process above. Control firms experience no productivity change. Sales are realized at the firm–market level and aggregated to the firm level.

Jensen's inequality. All firms operate in five markets in both periods. Firm–market sales are independent draws from a Gamma distribution,

$$S_{imt} \sim \Gamma(k_{it}, \lambda),$$

where k_{it} is the shape parameter and λ the common rate parameter. Control firms have shape k_C in both periods, while treated firms have shape k_C in period 1 and k_T in period 2. We vary k_T while holding $k_C = \lambda = 1$. The two groups have the same sales distribution when $k_T = k_C$. Relative to the exponential benchmark, this allows the variance of log growth to differ between treated and control firms. Under the exponential baseline, the variance of log growth is $\pi^2/3$ for both groups.

Entry and exit. We use the common data-generating process and vary η, the sensitivity of market participation to productivity. At $\eta = 0$, market participation is independent of productivity. Larger values increase the response of entry and exit to the treatment-induced productivity gain.

E.1.3 Confounding market shocks

We introduce market-level shocks that are correlated with treatment status. Specifically, each market j is subject to a multiplicative shock in the second period, so that firm–market sales satisfy

$$S_{ij,2} = X_{ij,2} \times \mathit{Shock}_j, \qquad \log(\mathit{Shock}_j) \sim N(0, 1).$$

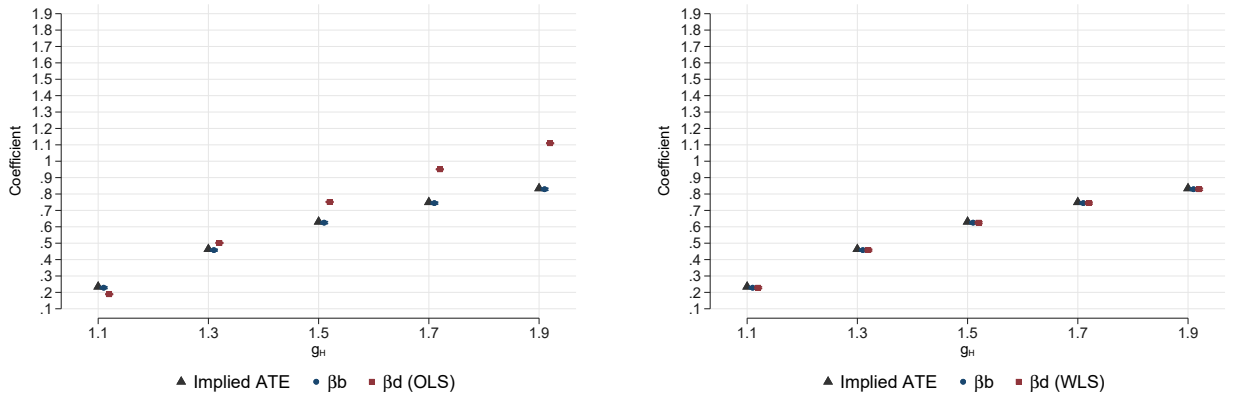
In this setting, the log-growth specification admits an additive decomposition, so that market fixed effects can absorb the confounding shock.

Treatment is assigned at the firm level as a function of firms' average pre-period exposure to these market shocks, so that treated firms are more likely to operate in markets with systematically different realizations of Shock_j. However, conditional on market fixed effects, treatment remains as good as randomly assigned across firms, since treatment depends only on market-level shocks, which are fully absorbed by the fixed effects, and not on firm–market residual shocks.

We set $\eta = 3$ and use a correlation parameter of -0.3, so treated firms are more exposed to negative market shocks, as in Table 2.

E.2 Additional disaggregation comparisons

Figure E.1: Disaggregation invariance: Implicit weights

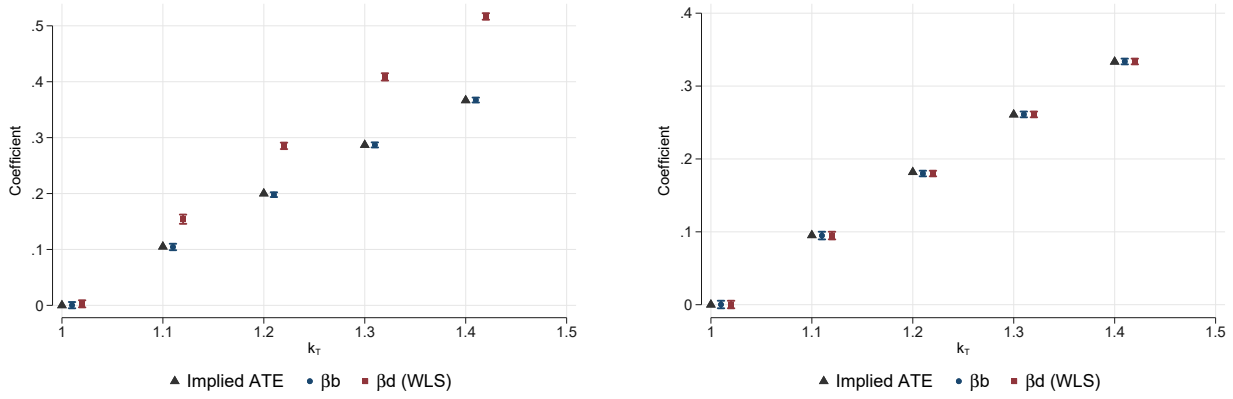


(a) Midpoint growth rate – OLS

(b) Midpoint growth rate – WLS

Note. The simulation follows Table E.1. Firms draw low or high initial productivity with equal probability. Low-productivity firms have $\theta_L = 0.1$, and high-productivity firms have $\theta_H = 3$. Pre-period productivity determines the number of markets, with $\eta = 3$. The number of markets is then held fixed across periods. Treated low-productivity firms receive productivity multiplier $g_L = 1.1$. Treated high-productivity firms receive the multiplier g_H shown on the x-axis. Control firms receive no productivity change.

Figure E.2: Disaggregation invariance: Jensen's inequality

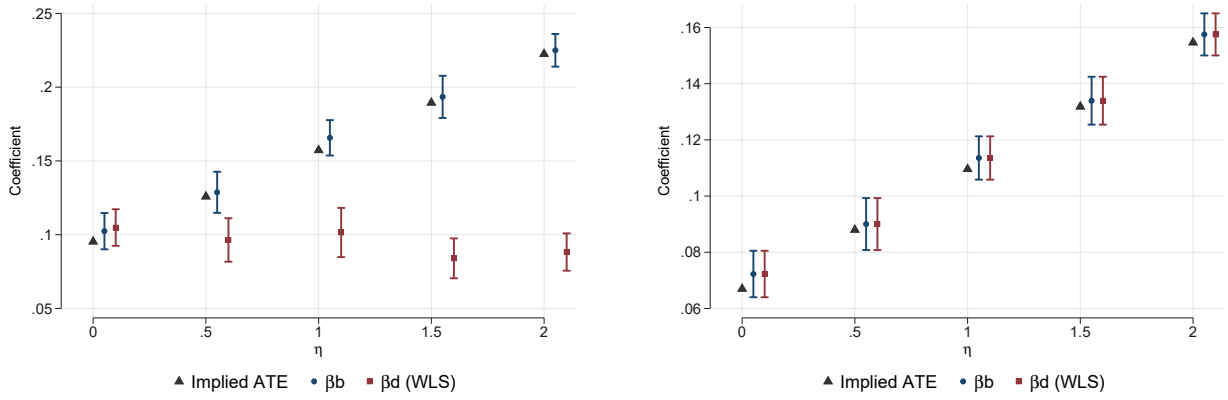


(a) Log growth rate – WLS

(b) Midpoint growth rate – WLS

Note. The simulation follows Table E.1. All firms operate in five markets in both periods. Firm–market sales are independent Gamma draws with shape k_{it} and common rate λ. Control firms keep shape k_C in both periods. Treated firms start from the same distribution as controls, but in period 2 their sales are drawn with shape k_T, shown on the x-axis. We hold $k_C = \lambda = 1$. The two groups have the same distribution when $k_T = k_C$.

Figure E.3: Disaggregation invariance: entry and exit

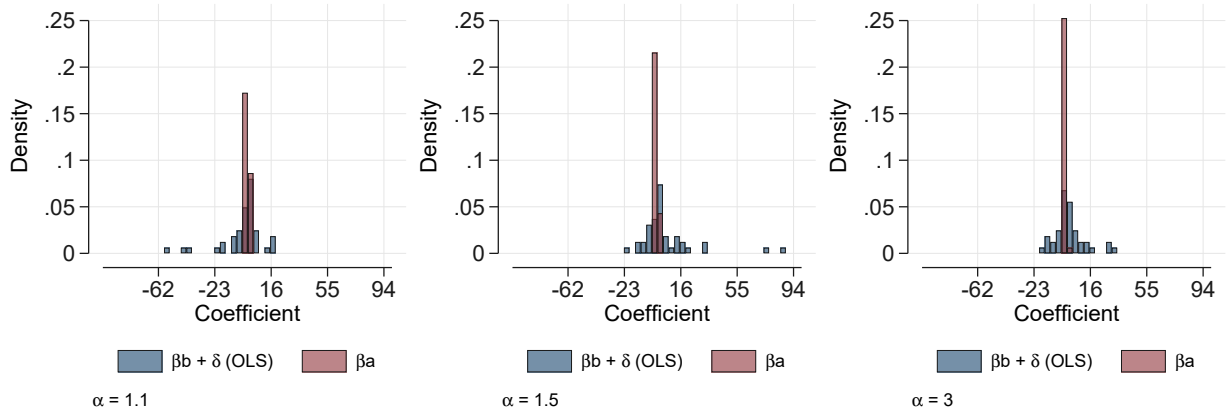


(a) Log growth rate – WLS

(b) Midpoint growth rate – WLS

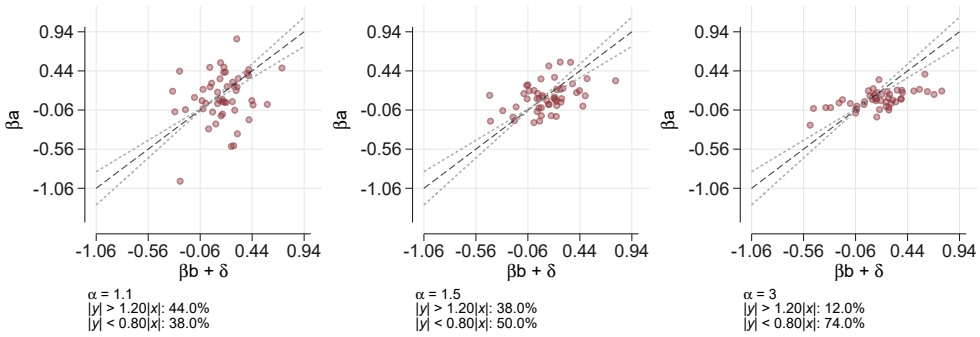
Note. The simulation follows Table E.1. The parameter η, which governs how strongly market participation responds to productivity, varies on the x-axis.

Figure E.4: Aggregation: Outliers in implicit weights simulation

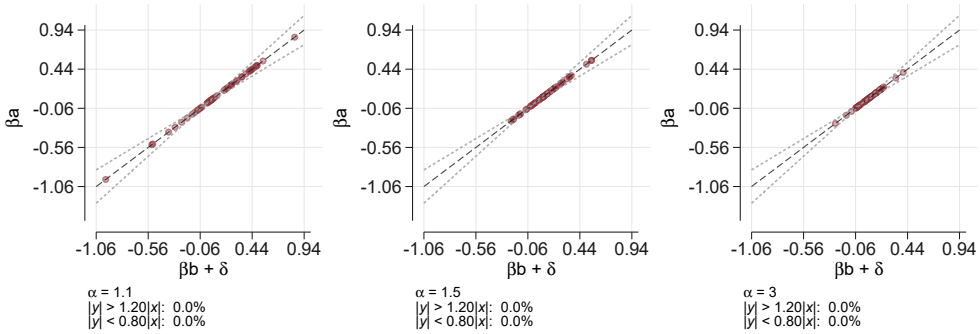


Note. Each figure shows the distribution of the coefficients obtained in 50 simulations for different values of α. Firm- and industry-level regressions use OLS regressions with standard growth rates.

Figure E.5: Aggregation invariance: Implicit weights



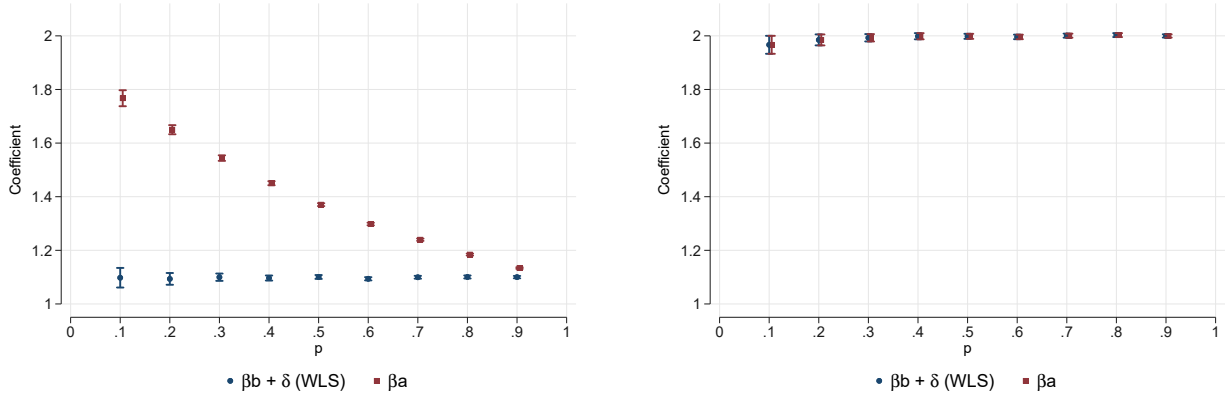
(a) Standard growth rate – OLS



(b) Standard growth rate – WLS

Note. The figure plots $\beta^b + \delta$ on the x-axis and β^a on the y-axis across simulations for different values of α. The simulation follows Table F.1. Initial firm productivity follows a Pareto distribution, $\theta_{i1} \sim \text{Pareto}(\alpha, \theta_{\min})$. We rescale $\theta_{\min}$ so that average productivity is fixed across values of α. Lower values of α generate more concentrated industries. Each point is one simulation. The dashed lines mark values of β^a that are 10% above or below $|\beta^b + \delta|$. In both panels, we exclude firms in the top 1% of sales growth rate.

Figure E.6: Aggregation invariance: Jensen's inequality



(a) Log growth rate – WLS

(b) Standard growth rate – WLS

Note. The simulation follows Table F.1. Each industry's treatment share is drawn with equal probability from either 0.01 or p_{high}, where p_{high} varies on the x-axis. For treated firms, productivity triples ($g = 2$). Each industry contains 1,000 firms, and there is no firm entry. Bars show 95% confidence intervals for mean estimates across 50 simulation repetitions.

F Aggregation simulations

This appendix reports the simulation setup and figures summarized in Section 4.4. The figures compare the industry coefficient with the firm-level combination from the exposure-augmented regression. The main comparisons use log growth with OLS and standard growth with matched WLS. The additional comparisons in Appendix E separate weighting from the growth-rate transformation.

F.1 Data-generating process

These simulations quantify the gap between firm-level and industry-level estimates under standard empirical specifications. Our data-generating process mirrors the one described in Section 3.3.1, with three key differences. First, firms no longer operate across multiple markets. The exponential process used to model firm–market sales is therefore applied directly to total firm sales. Second, firms are assigned to industries, which differ in their exposure to treatment through the share of treated firms, drawn from a uniform distribution. Third, at time 2, new firms enter industries with probability p_{entry}. We set the number of potential entrants to 50. Entrants are assigned the same productivity as incumbent firms at time 1. For simplicity, we do not model firm exit. Table F.1 reports the baseline parameters used in the aggregation simulations.

Table F.1: Aggregation simulation: baseline parameters

Parameter	Description	Default
Initial firm heterogeneity		
θ_1	Initial productivity	1
Firm sales: $S \sim \text{Exponential}(\mu_0\, \theta^\gamma)$		
μ_0	Baseline sales multiplier	1
γ	Sensitivity to productivity	1
Treatment: $\theta_2 = (1+g) \cdot \theta_1$		
g_{control}	Productivity upgrade for control firms	0
g_{treated}	Productivity upgrade for treated firms	0.1
Industry-level share of treated firms		
Share Treated$_j$	Probability that firm i within industry j is treated	U(0.1, 0.9)
Entry $p_{\text{entry}} = \kappa \cdot$ *Share Treated*$_j$		
N_e	Number of potential entrants	50
κ	Sensitivity of entry probability to industry treatment share	0.4
θ_e	Productivity of new entrants	1
Simulation specifications		
N_{ind}	Number of industries per simulation	100
$N_{\text{firm/ind}}$	Number of firms per industry	100
N_{sim}	Number of simulations	50

The aggregation simulations compare standard growth with lagged-sales WLS weights to log growth estimated by OLS. The firm-level regression includes industry exposure in both cases.

Under log-growth OLS, its coefficient on firm-level treatment need not equal the sales-weighted fixed-effect baseline in the main text. These figures retain that common-practice benchmark and compare its combination $\beta^b + \delta$ with the corresponding industry coefficient.

We also do not report implied average treatment effects in this section. The industry-level average treatment effect can be computed in a manner similar to Section 3.1. However, aggregation is typically used not to estimate an industry-level ATE per se, but to compare firm- and industry-level estimates as a way to detect potential SUTVA violations. For this reason, we view the comparison between aggregation levels, rather than the industry-level ATE, as the relevant object.

F.2 Implicit weights

We now show how differences in implicit weighting across levels of aggregation can break invariance. Throughout, we compare the industry-level coefficient β^a, obtained from the regression of industry growth on $Share\,Treated_j$, to the firm-level counterpart $\beta^b + \delta$, obtained from the augmented firm-level specification in Section 4.3. Implicit weights may lead β^a and $\beta^b + \delta$ to diverge for two distinct reasons.

First, when using OLS, more populated industries receive a larger weight in firm-level regressions. As a result, firm-level estimates may differ from industry-level ones if industries with more firms respond differently to treatment. This mechanism is identical to the one discussed in Section 3.3.2, so we do not elaborate further here.

Second, aggregation introduces an additional source of weighting that is specific to the industry level. At the firm level, an unweighted regression assigns equal importance to each firm. By contrast, industry-level outcomes aggregate firm sales, so firms with larger sales contribute more to industry growth. Industry regressions therefore implicitly weight firms by their size.

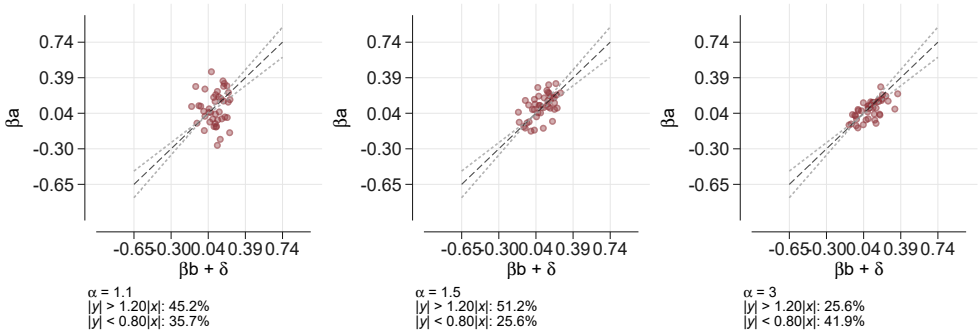
This implicit weighting creates a wedge between firm- and industry-level estimates whenever large firms behave differently, either because of heterogeneous treatment effects or because they are subject to idiosyncratic shocks whose effects on sales levels scale with firm size. In concentrated industries, where a small number of firms account for a large share of industry activity, this mechanism can be quantitatively important, as industry-level estimates primarily reflect the behavior of these few dominant firms rather than that of the average firm.

We illustrate this mechanism using the following simulation. We focus on the intensive margin by assuming that all firms are present at both periods. We maintain the assumption of homogeneous treatment and instead focus on the role of idiosyncratic shocks. Because sales are drawn from an exponential distribution, realized growth rates already contain idiosyncratic variation across firms. To capture differences in industry concentration, we draw initial firm productivity from a Pareto distribution,

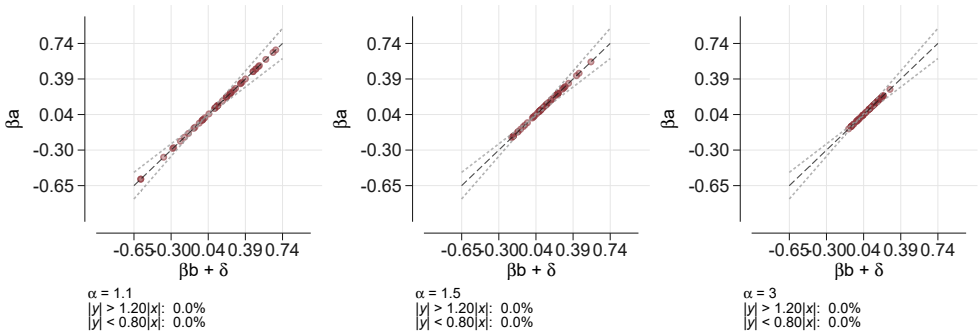
$$\theta_{i,t_1} \sim \text{Pareto}(\alpha, \theta_{\min}),$$

where the tail index α governs concentration. Smaller values of α generate a heavier right tail, leading to highly concentrated industries in which a few firms dominate sales. As α increases, the distribution becomes less skewed and firms are more similar in size. We rescale the lower bound $\theta_{\min}$ so that average productivity is held fixed across specifications.

Figure F.1: Aggregation invariance: Implicit weights



(a) Log growth rate – OLS



(b) Standard growth rate – WLS

Note. The figure plots $\beta^b + \delta$ on the x-axis and β^a on the y-axis across simulations for different values of α. The simulation follows Table F.1. Initial firm productivity follows a Pareto distribution, $\theta_{i1} \sim \text{Pareto}(\alpha, \theta_{\min})$. We rescale $\theta_{\min}$ so that average productivity is fixed across values of α. Lower values of α generate more concentrated industries. Each point is one simulation. The dashed lines mark values of β^a that are 10% above or below $|\beta^b + \delta|$.

Figure F.1 reports $\beta^b + \delta$ on the x-axis and β^a on the y-axis across simulations for different values of α. Panel (a) uses log growth rates estimated by OLS, while panel (b) uses standard growth rates estimated by WLS. To quantify the discrepancy between firm- and industry-level estimates, we report below each figure the percentage of simulations in which β^a and $\beta^b + \delta$ differ by more than 10%, corresponding to the points outside the dashed lines.[2]

The finite-sample contribution of idiosyncratic shocks is distinct from the population discrepancy due to heterogeneous treatment effects. Under homogeneous effects, exposure-independent shocks, and regularity conditions ensuring convergence across independent industries, chance covariance with those shocks vanishes as the number of industries grows. This does not imply that the Jensen component of the log-growth comparison disappears.

Panel (a) shows that for low values of α, corresponding to highly concentrated industries, β^a

[2] Figure E.5 compares standard growth rates estimates using OLS and WLS and shows that using weighted least squares is necessary to achieve aggregation invariance. Relative to log growth rates with OLS, standard growth rates with OLS raise an additional issue: they can generate extreme outliers when sales in the first period are very small. Weighted least squares addresses this problem mechanically, since observations with small initial sales receive small weights. With OLS, however, these outliers can generate extremely large coefficients, as shown in Figure E.4. For illustration, we therefore exclude firms whose sales growth rate is in the top 1% and recompute industry aggregates accordingly.

and $\beta^b + \delta$ diverge substantially, even though SUTVA holds. Because this divergence is driven by idiosyncratic shocks affecting large firms, the sign of the difference between the coefficients can be either positive or negative. Panel (b) shows that this discrepancy disappears when using the standard growth rate with WLS. In this case, the firm-level regression assigns the same weights as the industry-level regression, so $\beta^a = \beta^b + \delta$ in all simulations and for all values of α.

F.3 Jensen inequality

As when disaggregating, industry-level growth is not a linear aggregation of firm-level growth rates when using log growth rates. Similarly to Section 3.3.3, we define the concavity term as the difference between industry log growth and the weighted average of firm-level log growth rates:

$$C_{jt} \equiv \log\left(\sum_{i:\, j(i)=j} \omega_{ij}^s \exp(\Delta^l S_{it})\right) - \sum_{i:\, j(i)=j} \omega_{ij}^s \Delta^l S_{it}.$$

Applying the same second-order approximation as in equation (9), we obtain that the concavity term depends on the within-industry variance of firm growth rates. Characterizing the conditions under which aggregation invariance holds, however, is more complex than in the disaggregation case because variation in treatment status within industries directly contributes to the within-industry variance of firm growth rates. To characterize the role of treatment exposure, we use equation (2) to express firm growth rates as the sum of a treatment component and a residual.[3]

Under random assignment of treatment within industries, the covariance between treatment status and residual firm growth vanishes in the population. Ignoring higher-order terms, the concavity term can therefore be written as

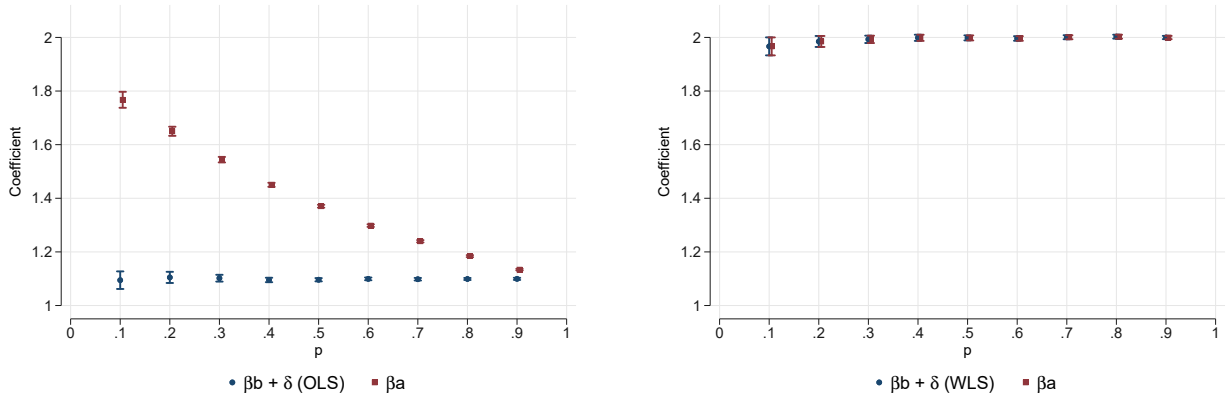
$$C_{jt} \approx \frac{1}{2}\left(\beta^2 Share\,Treated_j(1 - Share\,Treated_j) + \mathrm{Var}_{\omega,j}(u_{ijt})\right).$$

The first term reflects the variation due to differences in treatment status across firms within an industry. Because treatment is binary, this component depends on the share of treated firms and is largest when treatment exposure is evenly distributed within the industry. The second term reflects the residual dispersion of firm growth rates within industries. This second component is the direct counterpart of the Jensen mechanism emphasized in the disaggregation setting, where differences in within-firm dispersion of firm–market growth rates lead firm- and firm–market regressions to identify different estimands. We focus therefore on the first term in the following empirical analysis.

When will within-industry variability in treatment status lead aggregation invariance to break? Since industry log growth can be written as the sum of the weighted average of firm-level log growth rates and the concavity term, the industry-level coefficient absorbs both the average effect of treatment on firm growth and the way treatment exposure mechanically affects the dispersion of firm growth rates within industries through the concavity term. Aggregation invariance under log growth therefore requires the concavity term to be orthogonal to $Share\,Treated_j$.

[3] We do not include the treatment share in the computation of the concavity term since aggregation invariance is defined conditional on the absence of SUTVA violations.

Figure F.2: Aggregation invariance: Jensen's inequality



(a) Log growth rate – OLS

(b) Standard growth rate – WLS

Note. The simulation follows Table F.1. Each industry's treatment share is drawn with equal probability from either 0.01 or p_{high}, where p_{high} varies on the x-axis. For treated firms, productivity triples ($g = 2$). Each industry contains 1,000 firms, and there is no firm entry. Bars show 95% confidence intervals for mean estimates across 50 simulation repetitions.

For two exposure values, p_{low} and p_{high}, holding residual dispersion fixed, the treatment-status component contributes the following slope to the industry regression:

$$\frac{C(p_{high}) - C(p_{low})}{p_{high} - p_{low}} \approx \frac{\beta^2}{2}(1 - p_{low} - p_{high}).$$

Its magnitude is greatest when both treatment shares are close to zero or both are close to one, with opposite signs in the two cases. Thus, a narrow range of treatment exposure near either endpoint can generate a large coefficient discrepancy even though the concavity terms themselves are small.

We illustrate this mechanism by assigning industries one of two treatment shares with equal probability: $p_{low} = 0.01$ or p_{high}, which varies over $[0.1, 0.9]$. Almost all firms are untreated in low-exposure industries. For p_{high} below one half, high-exposure industries have greater within-industry variation in treatment status and therefore a larger concavity term.

Above one half, further increases in p_{high} reduce within-industry treatment variation and the concavity term. With $p_{low} = 0.01$, the slope above is zero at $p_{high} = 0.99$. The coefficient discrepancy therefore becomes small as p_{high} approaches one.

Figure F.2 displays the simulation results.[4] There is no entry. We use a large treatment effect ($g = 2$) to make the concavity term quantitatively relevant, since it is proportional to β^2.[5] Panel (a) uses log growth rates estimated by OLS, while panel (b) uses standard growth rates estimated by WLS. As expected, the discrepancy between the firm- and industry-level estimates when using log growth rates is largest for low values of p_{high}, while aggregation invariance nearly holds when p_{high} approaches one. Panel (b) shows that aggregation invariance always holds when using the standard growth rate with WLS.

[4] Figure E.6 compares log and standard growth rates estimates using WLS and shows that the standard growth rate is necessary to achieve aggregation invariance.

[5] We use 1,000 firms per industry to estimate treatment effects precisely even when $p_{low} = 0.01$.

F.4 Entry and exit

This simulation, reported in Table 4, combines treatment-induced firm entry with negative spillovers on incumbent firms. It follows the common framework in Table F.1, with the following parameter choices. Treated firms receive a productivity multiplier $H_{treated} = 1.1$. We introduce $\bar{N} = 50$ potential entrant firms per industry. Each potential entrant enters in period t_2 with probability

$$p_{\text{entry},j} = \kappa \times ShareTreated_j, \qquad \kappa = 0.375.$$

Industries with a larger share of treated firms therefore experience more entry, increasing industry growth. This value of κ overrides the default in Table F.1 for this exercise.

At the same time, incumbent firms are exposed to a negative intensive-margin spillover. In period t_2, incumbent sales are multiplied by

$$\phi_j = 1 - \gamma \times ShareTreated_j, \qquad \gamma = 0.25.$$

A larger share of treated firms therefore reduces incumbent sales growth within the industry. As in the common framework, we do not model firm exit. The table reports means and standard errors across 50 simulations.

G Details of the military-procurement application

G.1 Variables and estimation

We combine annual real net military procurement, county population, and county–industry employment. We measure spending using real net contract obligations, excluding construction. Industry employment cells are based on County Business Patterns and reconciled to county totals, including residual industry cells. The retained support is 3,058 county groups. The three-year regressions cover 1981–2016.

As in Section 3.5, let Spending_{ct}, Pop_{ct} and E_{ct} denote procurement, population and employment. An overbar denotes average annual procurement over the three years ending in t. Writing x_{ct} for $\text{County spending}_{ct}$, county spending before winsorization is

$$x_{ct}^{raw} = \frac{\overline{\text{Spending}}_{ct} - \text{Spending}_{c,t-3}}{\text{Pop}_{c,t-3}}.$$

We winsorize this spending variable once at the pooled first and ninety-ninth percentiles. Employment growth and the instrument are not winsorized. Historical procurement exposure and the national shift are

$$a_{ct} = \frac{\sum_{r\in\mathcal{H}_t} \text{Spending}_{cr}}{\sum_{r\in\mathcal{H}_t} \text{Pop}_{cr}}, \qquad n_t = \frac{\sum_c (\overline{\text{Spending}}_{ct} - \text{Spending}_{c,t-3})}{\sum_c \text{Pop}_{c,t-3}},$$

where $\mathcal{H}_t$ is the five-year historical window from $t-12$ to $t-8$. The excluded county instrument is $z_{ct} = a_{ct} n_t$. National sums use the retained county support and include the focal county.

Disaggregation repeats x_{ct} and z_{ct} across industries. Log weights are lagged industry employment divided by county employment, without survivor renormalization. Midpoint weights use industry employment at both endpoints divided by county employment at both endpoints. For aggregation to region s, the county weights, regional exposure and regional instrument are

$$\omega_{cst} = \frac{E_{c,t-3}}{\sum_{k\in s} E_{k,t-3}}, \qquad X_{st} = \sum_{c\in s} \omega_{cst} x_{ct}, \qquad Z_{st} = \sum_{c\in s} \omega_{cst} z_{ct}.$$

Here X_{st} is $\text{Regional exposure}_{st}$ in the main text. Regional exposure thus uses the weights that aggregate employment growth. It is not reconstructed as total regional procurement divided by regional population. Separately estimated region regressions have one observation per region–year and equal weights. With time-varying employment shares, the weighted average of county fixed effects need not be absorbed by region fixed effects. The coefficient relationship can therefore have a small residual, as documented in Section 4.6.

G.2 Covariate balance and treatment heterogeneity

As a preliminary diagnostic, we adapt equation (30) to continuous county spending and IV. For each county characteristic C_c measured in 1978, we estimate

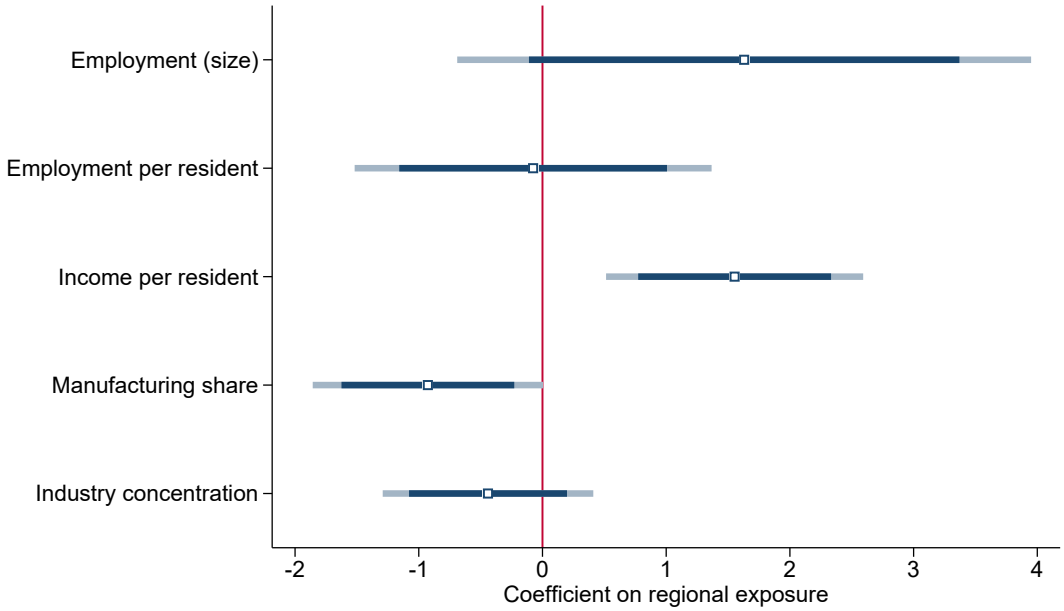
$$x_{ct}C_c = \alpha_c + \lambda_t + \rho_C x_{ct} + \pi_C X_{st} + u_{ct},$$

instrumenting x_{ct} and X_{st} with z_{ct} and Z_{st}. These initial diagnostics do not include size-group interactions or the other heterogeneity controls in equation (30). Figure G.1 reports the five characteristics, standardized across counties. Income per resident is winsorized at the national county first and ninety-ninth percentiles before standardization. We use lagged employment-share weights, county and year fixed effects, and state-clustered standard errors with a t_{50} reference distribution.

We then allow the response to spending to vary separately with employment in levels, manufacturing employment share, and income per resident. For each characteristic, the regression adds $x_{ct}C_c$, its weighted regional average $Q_{st} = \sum_c \omega_{cst} x_{ct} C_c$, and the regional characteristic mean $A_{st} = \sum_c \omega_{cst} C_c$. The two added excluded instruments are $z_{ct}C_c$ and $\sum_c \omega_{cst} z_{ct} C_c$. The county characteristic itself is absorbed by county fixed effects. These interaction instruments require the corresponding conditional exclusion restrictions.

Table G.1 reports exposure coefficients of 8.70, 10.18 and 8.33 after the separate employment, manufacturing and income adjustments, compared with 8.08 without these controls. Income increases the standard error to 7.99. These comparisons address observed sources of treatment heterogeneity; observable balance cannot establish the absence of unobserved heterogeneity.

Figure G.1: Covariate balance: initial county characteristics



Note. Points are $\widehat{\pi}_C$ in the balance regression above. Pale and dark bars show 99% and 95% confidence intervals. Characteristics are measured in 1978. The sample, weights, fixed effects and instruments match the initial state-aggregation specification. No 10^5 display multiplier applies to these coefficients.

Table G.1: Spillovers with separate treatment-heterogeneity controls

	No controls	Employment (size)	Manufacturing share	Income per resident
	δ	δ	δ	δ
	(1)	(2)	(3)	(4)
	8.076**	8.698**	10.180***	8.329
	(3.263)	(4.002)	(3.558)	(7.990)
p-value	0.017	0.035	0.006	0.302
First-stage F				
County spending	41.44	29.19	66.67	61.61
Regional exposure	41.86	28.17	62.87	72.87
Characteristic × county spending	–	64.73	51.22	138.35
Regional characteristic interaction	–	27.87	72.09	138.59
Observations	110,088	110,088	110,088	110,088
Clusters	51	51	51	51

Note. Only the regional-exposure coefficient is reported. Controls use 1978 characteristics and enter one at a time. The regional characteristic interaction is Q_{st}, not generally $X_{st}A_{st}$. Each ordinary first-stage F tests all excluded instruments jointly for the indicated endogenous variable. The controlled regressions have four endogenous variables and four excluded instruments. Coefficients and standard errors are multiplied by 10^5. Standard errors are clustered by state and p-values use t_{50}. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.